CLOUD COMPUTING

LAB NO# 05

SUBMITTED TO:

Sir Shohaib

SUBMITTED BY:

Name: Amina Noor

Reg No. 2023-BSE-007

Semester: V-A

CLOUD COMPUTING

LAB NO# 05

Task 1: Discover missing command & install Java using apt suggestion

Step 1: Check if Java is installed

Open your VM terminal and type:

Java

```
amina@2023-bse-007:~$ java
Command 'java' not found, but can be installed with:
sudo apt install openjdk-17-jre-headless  # version 17.0.17+10-1~24.04, or
sudo apt install openjdk-21-jre-headless  # version 21.0.9+10-1~24.04
sudo apt install default-jre              # version 2:1.17-75
sudo apt install openjdk-11-jre-headless  # version 11.0.29+7-1ubuntu1~24.04
sudo apt install openjdk-25-jre-headless  # version 25.0.1+8-1~24.04
sudo apt install openjdk-8-jre-headless   # version 8u472-ga-1~24.04
sudo apt install openjdk-19-jre-headless  # version 19.0.2+7-4
sudo apt install openjdk-20-jre-headless  # version 20.0.2+9-1
sudo apt install openjdk-22-jre-headless  # version 22~22ea-1
amina@2023-bse-007:~$ _
```

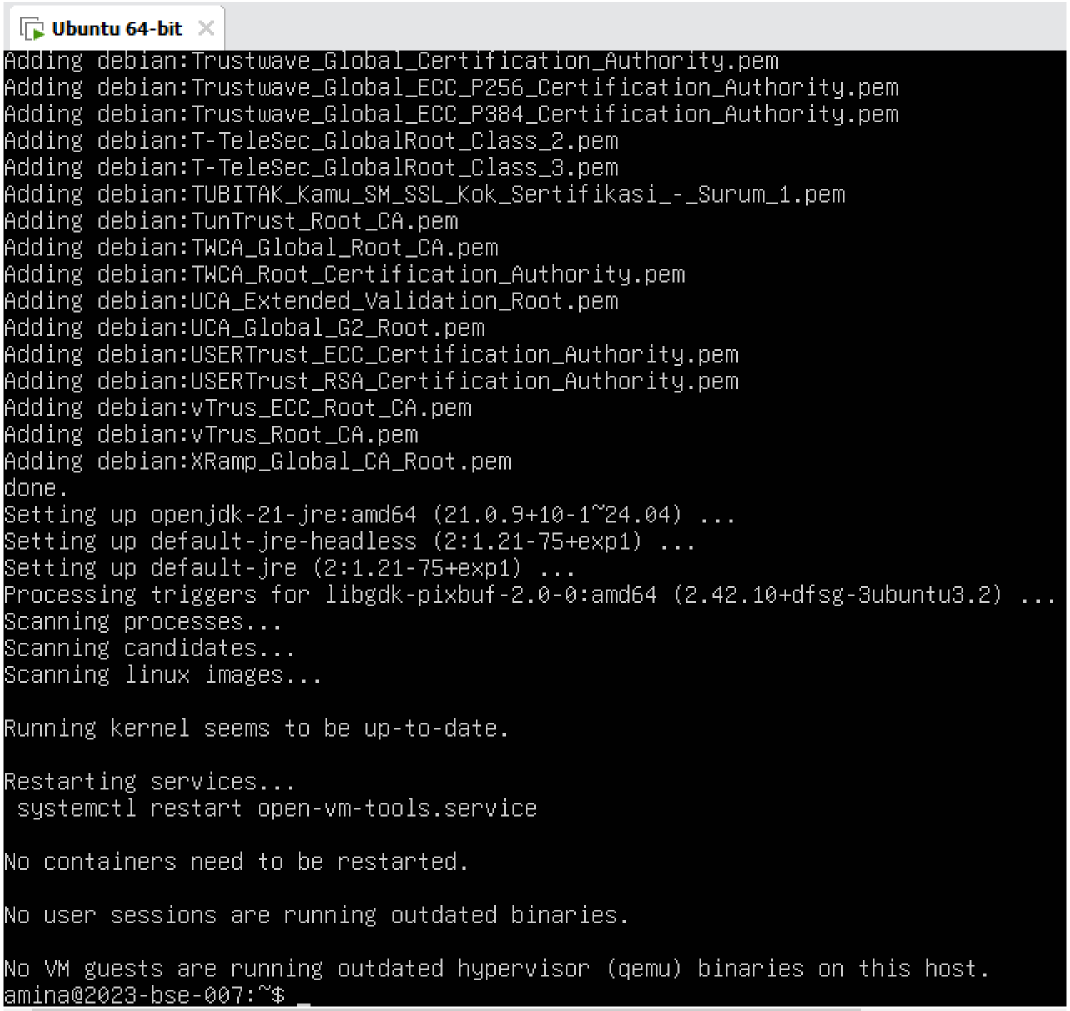
Step 2: Install Java using the suggested package

Use the **exact package name** shown on your screen.

Example (your package name may be different):

sudo apt install default-jre -y

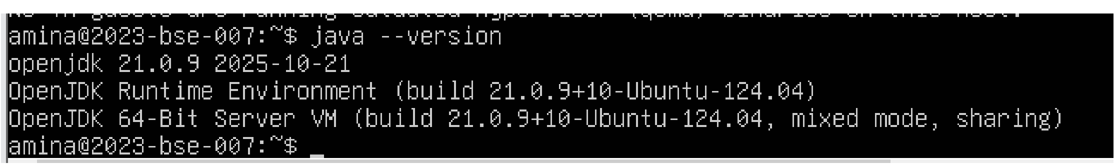
- Enter your **password** if asked
- Wait for installation to finish



Step 3: Verify Java installation

Run:

java --version

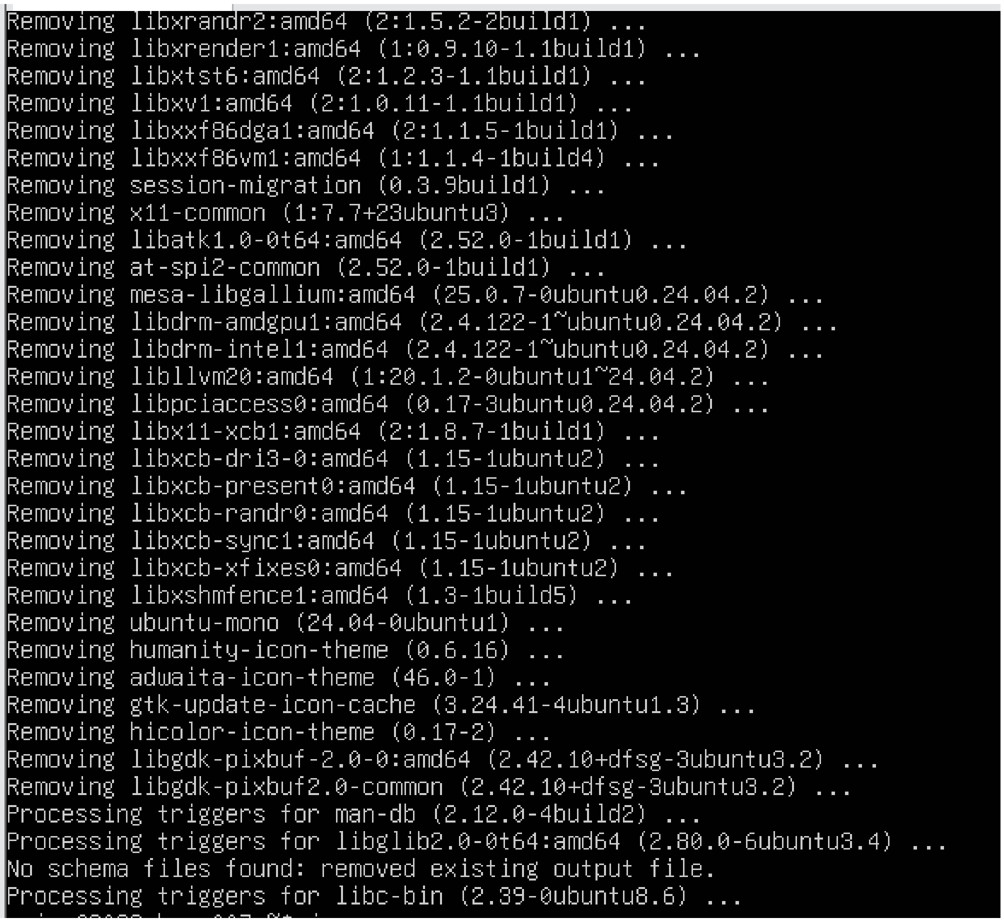


Step 4: Remove Java package

Remove the **same package you installed**:

Example:

sudo apt remove default-jre -y



```
Removing libxrandr2:amd64 (2:1.5.2-2build1) ...
Removing libxrender1:amd64 (1:0.9.10-1.1build1) ...
Removing libxtst6:amd64 (2:1.2.3-1.1build1) ...
Removing libxv1:amd64 (2:1.0.11-1.1build1) ...
Removing libxxf86dga1:amd64 (2:1.1.5-1build1) ...
Removing libxxf86vm1:amd64 (1:1.1.4-1build4) ...
Removing session-migration (0.3.9build1) ...
Removing x11-common (1:7.7+23ubuntu3) ...
Removing libatk1.0-0t64:amd64 (2.52.0-1build1) ...
Removing at-spi2-common (2.52.0-1build1) ...
Removing mesa-libgallium:amd64 (25.0.7-0ubuntu0.24.04.2) ...
Removing libdrm-amdgpu1:amd64 (2.4.122-1~ubuntu0.24.04.2) ...
Removing libdrm-intel1:amd64 (2.4.122-1~ubuntu0.24.04.2) ...
Removing libllvm20:amd64 (1:20.1.2-0ubuntu1~24.04.2) ...
Removing libpciaccess0:amd64 (0.17-3ubuntu0.24.04.2) ...
Removing libx11-xcb1:amd64 (2:1.8.7-1build1) ...
Removing libxcb-dri3-0:amd64 (1.15-1ubuntu2) ...
Removing libxcb-present0:amd64 (1.15-1ubuntu2) ...
Removing libxcb-randr0:amd64 (1.15-1ubuntu2) ...
Removing libxcb-sync1:amd64 (1.15-1ubuntu2) ...
Removing libxcb-xfixes0:amd64 (1.15-1ubuntu2) ...
Removing libxshmfence1:amd64 (1.3-1build5) ...
Removing ubuntu-mono (24.04-0ubuntu1) ...
Removing humanity-icon-theme (0.6.16) ...
Removing adwaita-icon-theme (46.0-1) ...
Removing gtk-update-icon-cache (3.24.41-4ubuntu1.3) ...
Removing hicolor-icon-theme (0.17-2) ...
Removing libgdk-pixbuf-2.0-0:amd64 (2.42.10+dfsg-3ubuntu3.2) ...
Removing libgdk-pixbuf2.0-common (2.42.10+dfsg-3ubuntu3.2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libglib2.0-0t64:amd64 (2.80.0-6ubuntu3.4) ...
No schema files found: removed existing output file.
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
```

Step 5: Confirm Java is no longer available

Run:

Java

```
amina@2023-bse-007:~$ java
-bash: /usr/bin/java: No such file or directory
amina@2023-bse-007:~$
```

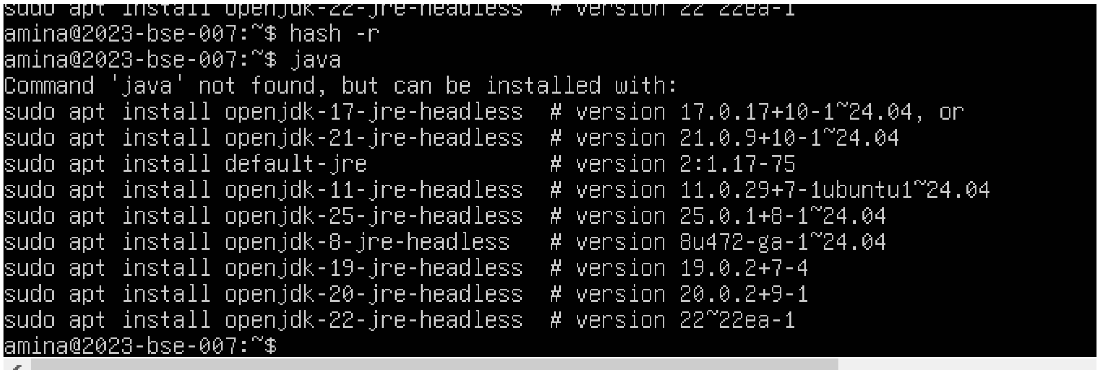
Step 6: Clear shell command cache (hash -r)

Run these **two commands together**:

hash -r

java

This clears the shell's cached command paths.



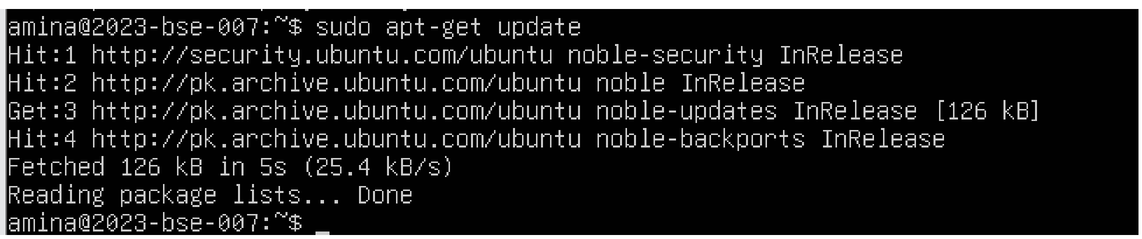
```
sudo apt install openjdk-22-jre-headless  # version 22~22ea-1
amina@2023-bse-007:~$ hash -r
amina@2023-bse-007:~$ java
Command 'java' not found, but can be installed with:
sudo apt install openjdk-17-jre-headless  # version 17.0.17+10-1~24.04, or
sudo apt install openjdk-21-jre-headless  # version 21.0.9+10-1~24.04
sudo apt install default-jre              # version 2:1.17-75
sudo apt install openjdk-11-jre-headless  # version 11.0.29+7-1ubuntu1~24.04
sudo apt install openjdk-25-jre-headless  # version 25.0.1+8-1~24.04
sudo apt install openjdk-8-jre-headless   # version 8u472-ga-1~24.04
sudo apt install openjdk-19-jre-headless  # version 19.0.2+7-4
sudo apt install openjdk-20-jre-headless  # version 20.0.2+9-1
sudo apt install openjdk-22-jre-headless  # version 22~22ea-1
amina@2023-bse-007:~$
```

Task 2: Install & Remove Java Using apt-get

Step 1: Update package list

Open your **VM terminal** and run:

sudo apt-get update

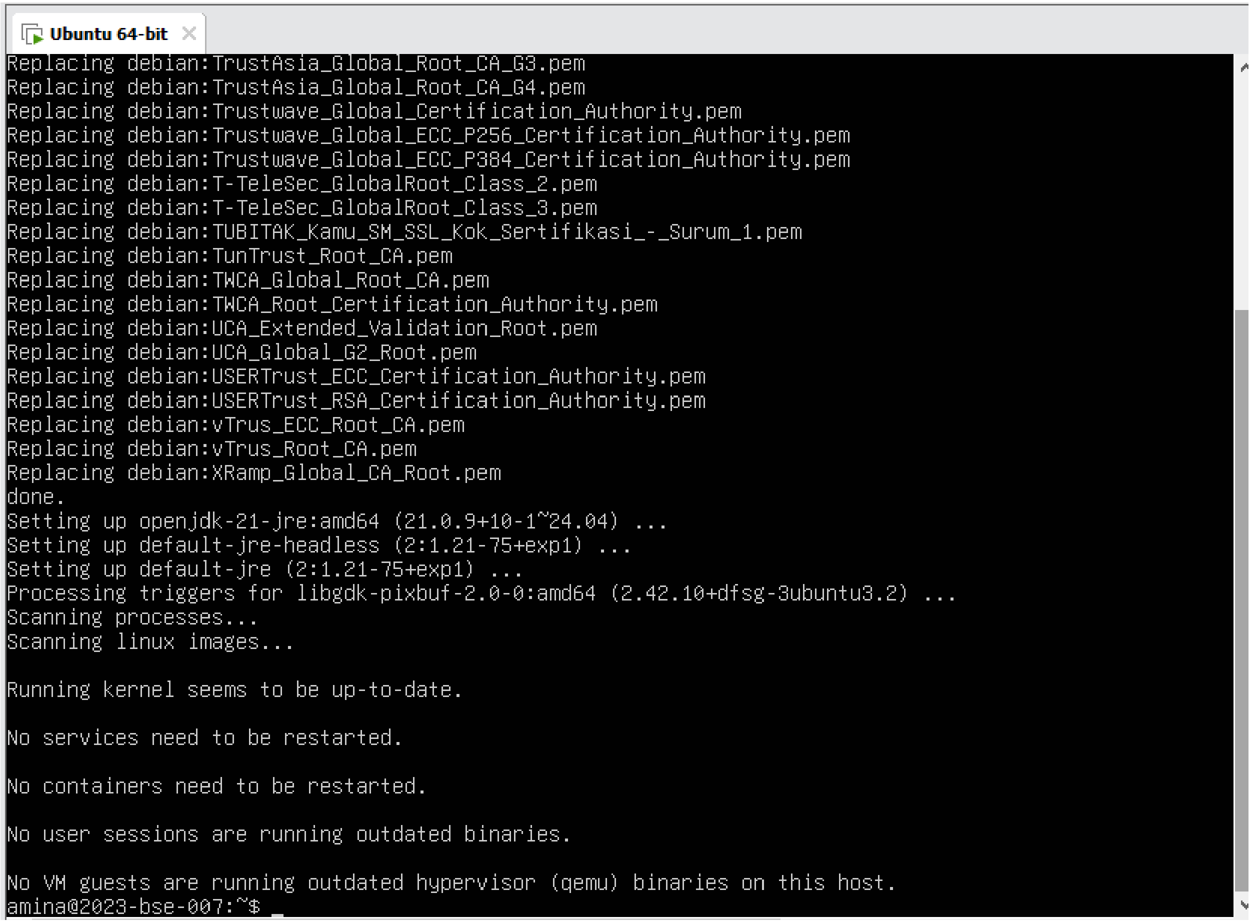


```
amina@2023-bse-007:~$ sudo apt-get update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://pk.archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://pk.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:4 http://pk.archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 126 kB in 5s (25.4 kB/s)
Reading package lists... Done
amina@2023-bse-007:~$ _
```

Step 2: Install Java using apt-get

Use the **same Java package** you used in Task 1
(usually default-jre)

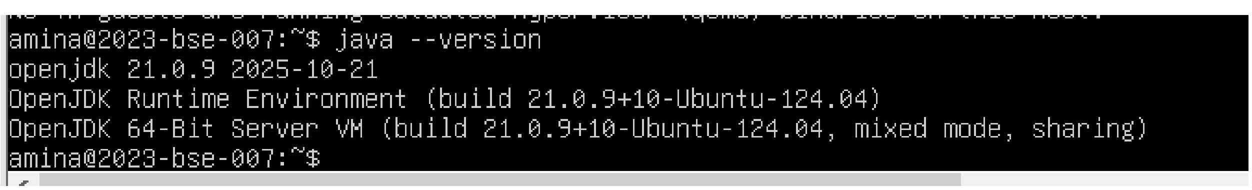
sudo apt-get install default-jre -y



Step 3: Verify Java installation

Run:

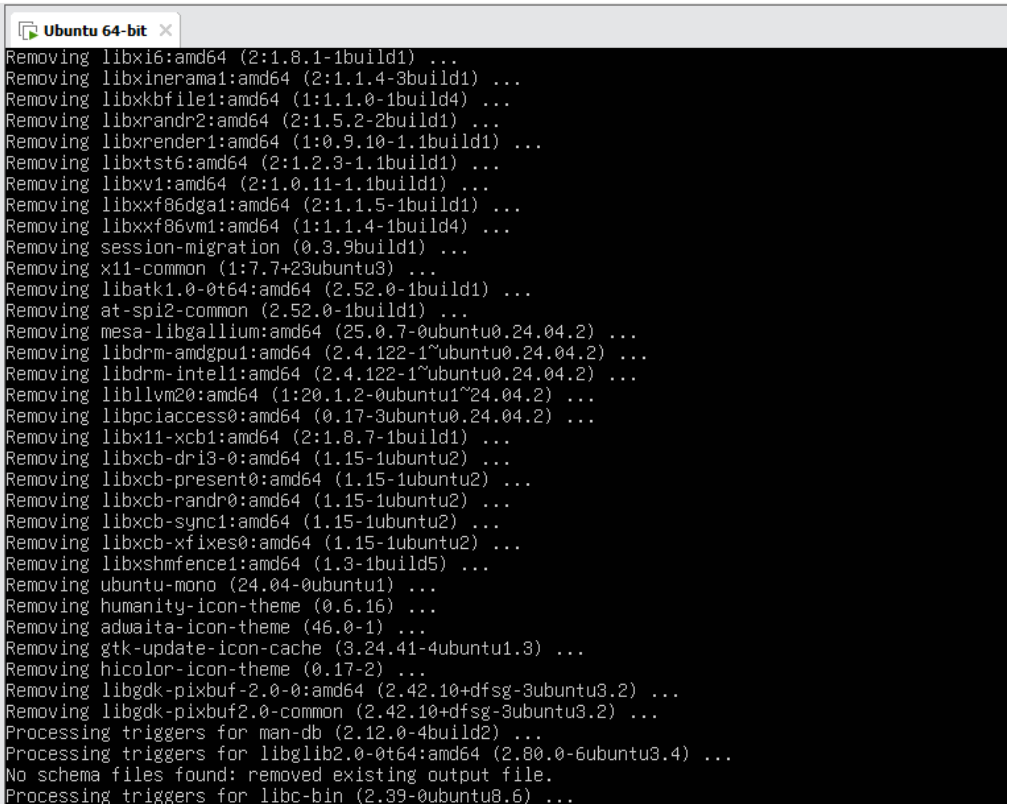
java --version



Step 4: Remove Java using apt-get

Remove the same package:

sudo apt-get remove default-jre -y

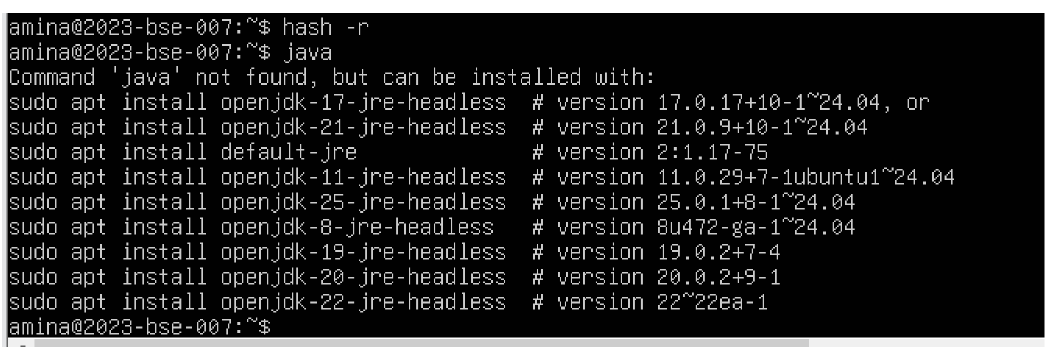


Step 5: Clear hash cache & confirm Java is removed

Run **both commands together**:

hash -r

java



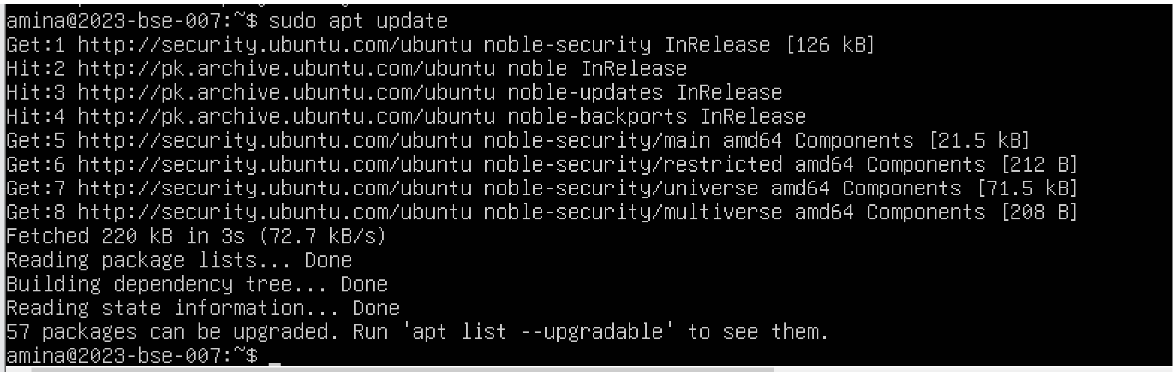
Task 3: apt update vs apt upgrade — Run & Explain

Step 1: Run apt update

This command **refreshes the package list** (it does NOT install anything).

In the terminal, run:

sudo apt update



Step 2: Run apt upgrade

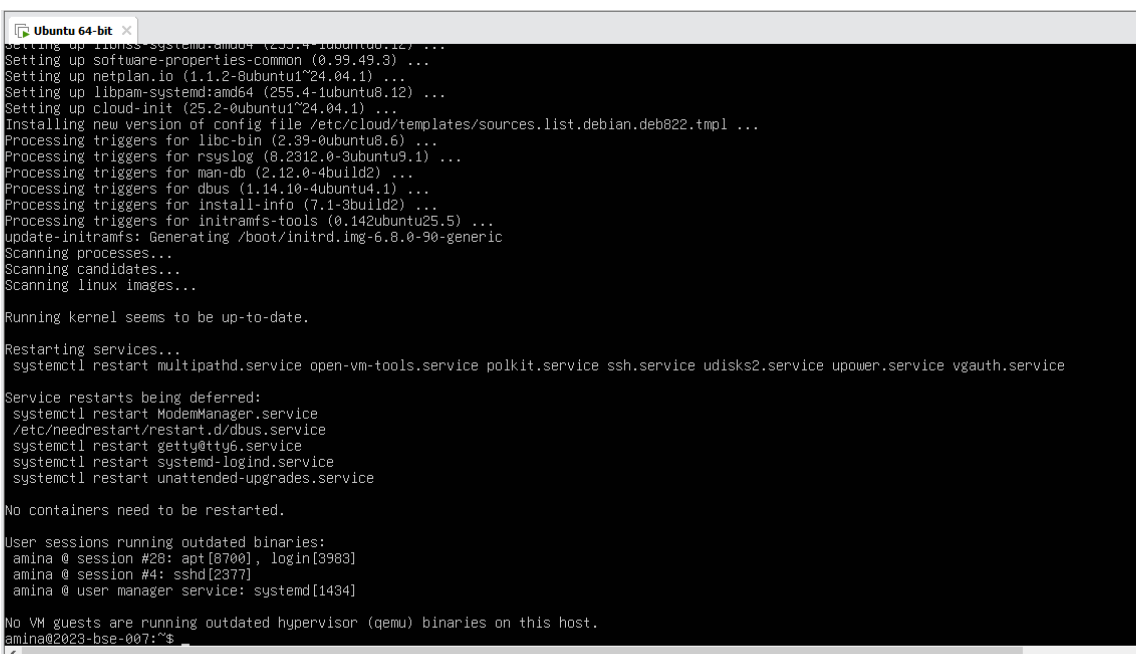
This command **installs newer versions** of already-installed packages.

Run:

sudo apt upgrade

- If it asks:
- Do you want to continue? [Y/n]

type **Y** and press **Enter**



Step 3: Write the explanation

Open a text file using nano:

nano ~/apt_update_vs_upgrade.md

Copy and paste this explanation:

The apt update command updates the local package index by downloading the latest information about available packages from repositories. The apt upgrade command uses the updated package index to install newer versions of packages that are already installed.

Save and exit nano:

- Press **CTRL + O → Enter**
- Press **CTRL + X**

Step 4: Display the file and take screenshot

Show the file contents:

cat ~/apt_update_vs_upgrade.md

Task 4: Install Visual Studio Code via snap (CLI-only)

Step 1: Install VS Code using snap

In your VM terminal, run:

sudo snap install --classic code

- Enter your password if asked
- Wait until installation completes
- You should see something like:
- code x.xx.x from Microsoft installed

Step 2: Verify VS Code is installed via snap

Run:

snap list code

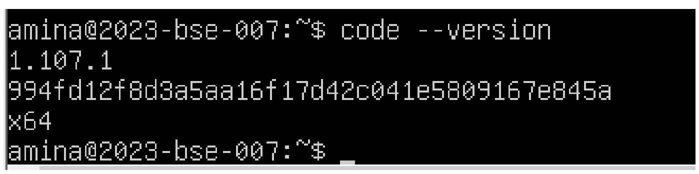
```
amina@2023-bse-007:~$ snap list code
Name  Version   Rev  Tracking       Publisher  Notes
code  994fd12f  217  latest/stable  vscode✓    classic
amina@2023-bse-007:~$ _
```

Step 3: Check VS Code version (CLI)

First try:

code --version

- On most systems this works even without GUI
- It prints version numbers and commit info

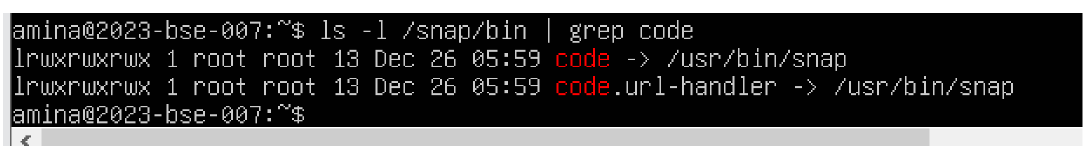


```
amina@2023-bse-007:~$ code --version
1.107.1
994fd12f8d3a5aa16f17d42c041e5809167e845a
x64
amina@2023-bse-007:~$ _
```

Step 4: Show where snap placed the code binary

Run:

ls -l /snap/bin | grep code



```
amina@2023-bse-007:~$ ls -l /snap/bin | grep code
lrwxrwxrwx 1 root root 13 Dec 26 05:59 code -> /usr/bin/snap
lrwxrwxrwx 1 root root 13 Dec 26 05:59 code.url-handler -> /usr/bin/snap
amina@2023-bse-007:~$
```

Task 5: Install XFCE GUI + XRDP and Launch VS Code

Step 1: SSH into your VM

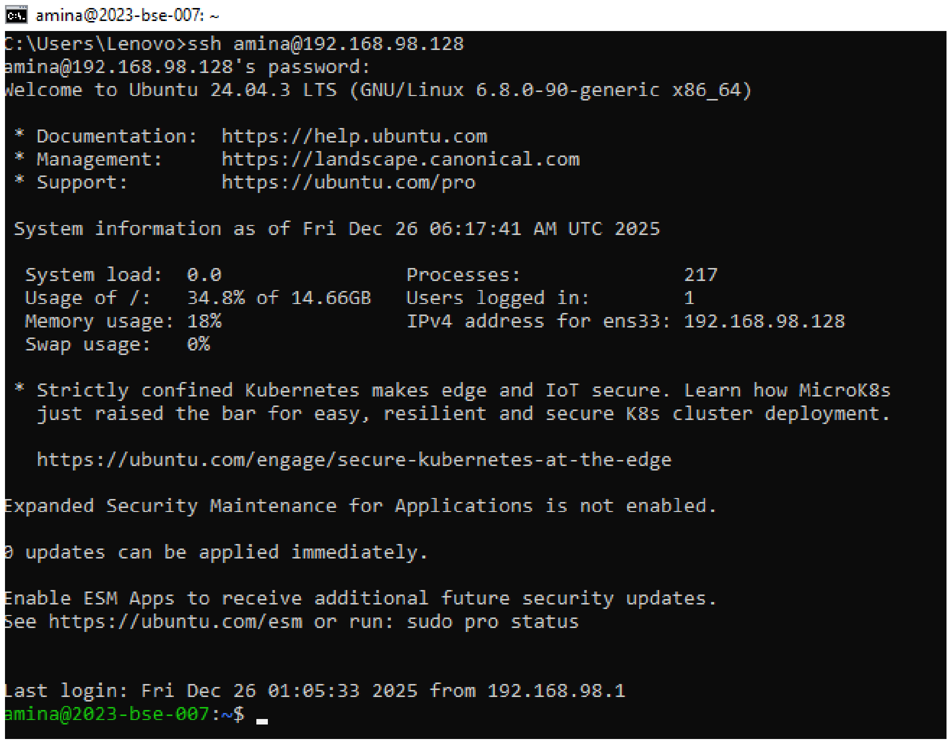
From your host system terminal (CMD):

ssh student@<VM-IP-ADDRESS>

Example:

ssh student@192.168.56.101

- Enter your VM password
- You should now see the Ubuntu server terminal



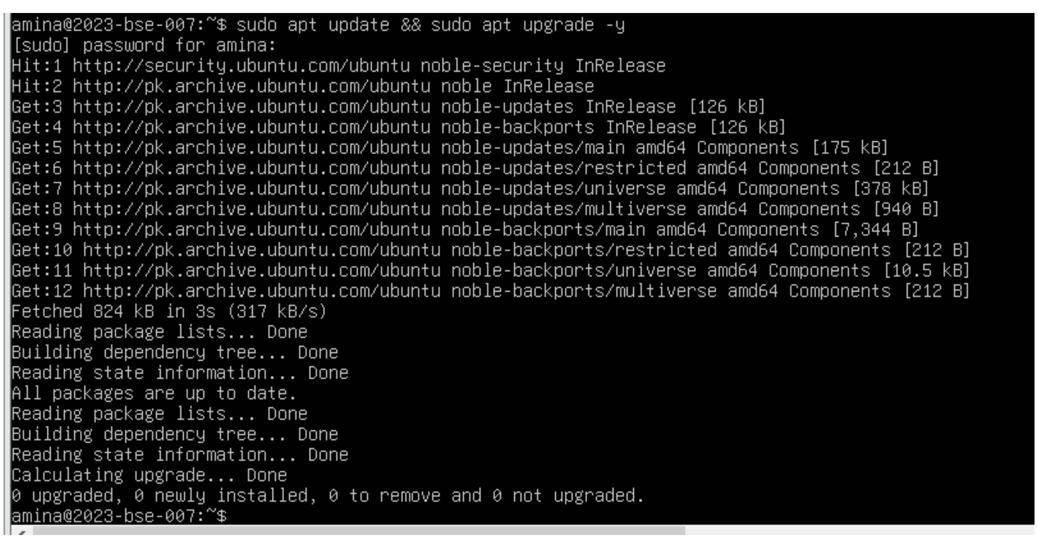
```
amina@2023-bse-007: ~
C:\Users\Lenovo>ssh amina@192.168.98.128
amina@192.168.98.128's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-90-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Fri Dec 26 06:17:41 AM UTC 2025

  System load:  0.0               Processes:             217
  Usage of /:   34.8% of 14.66GB  Users logged in:       1
  Memory usage: 18%               IPv4 address for ens33: 192.168.98.128
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Fri Dec 26 01:05:33 2025 from 192.168.98.1
amina@2023-bse-007:~$
```

Step 2: Update and upgrade the server

Run:

sudo apt update && sudo apt upgrade -y

- Let it finish fully



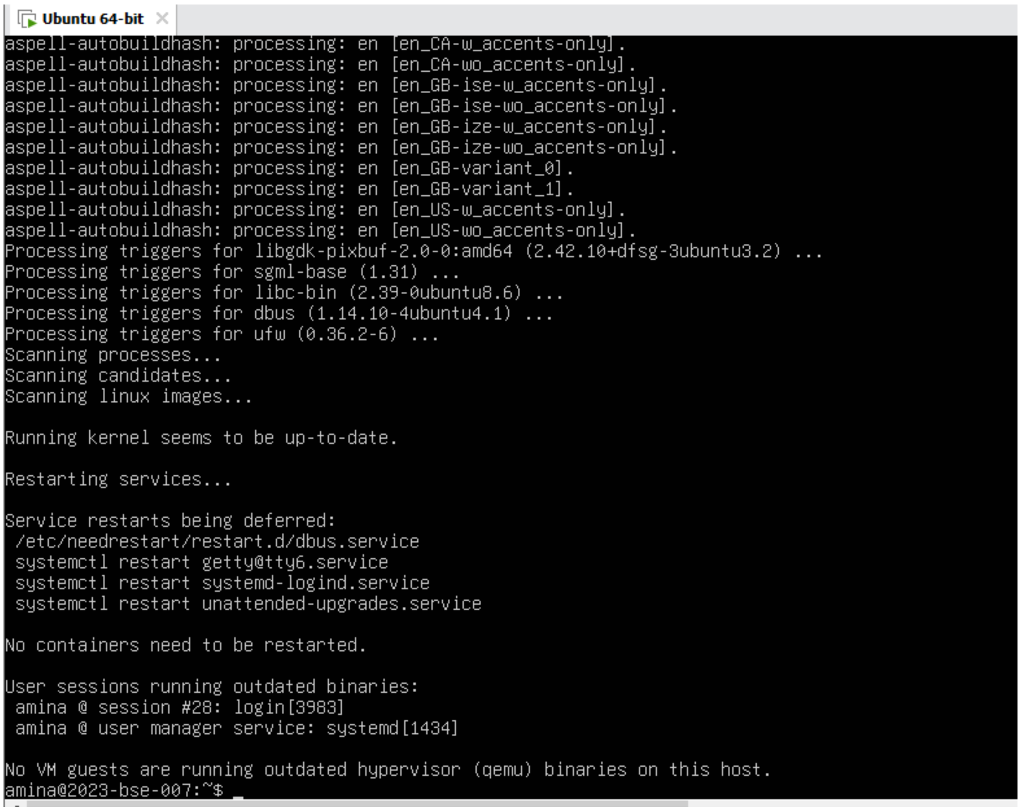
```
amina@2023-bse-007:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for amina:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://pk.archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://pk.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://pk.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:6 http://pk.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:7 http://pk.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
Get:8 http://pk.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:9 http://pk.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,344 B]
Get:10 http://pk.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:11 http://pk.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:12 http://pk.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 824 kB in 3s (317 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
amina@2023-bse-007:~$
```

Step 3: Install XFCE desktop (lightweight GUI)

Run:

sudo apt install xfce4 xfce4-goodies -y

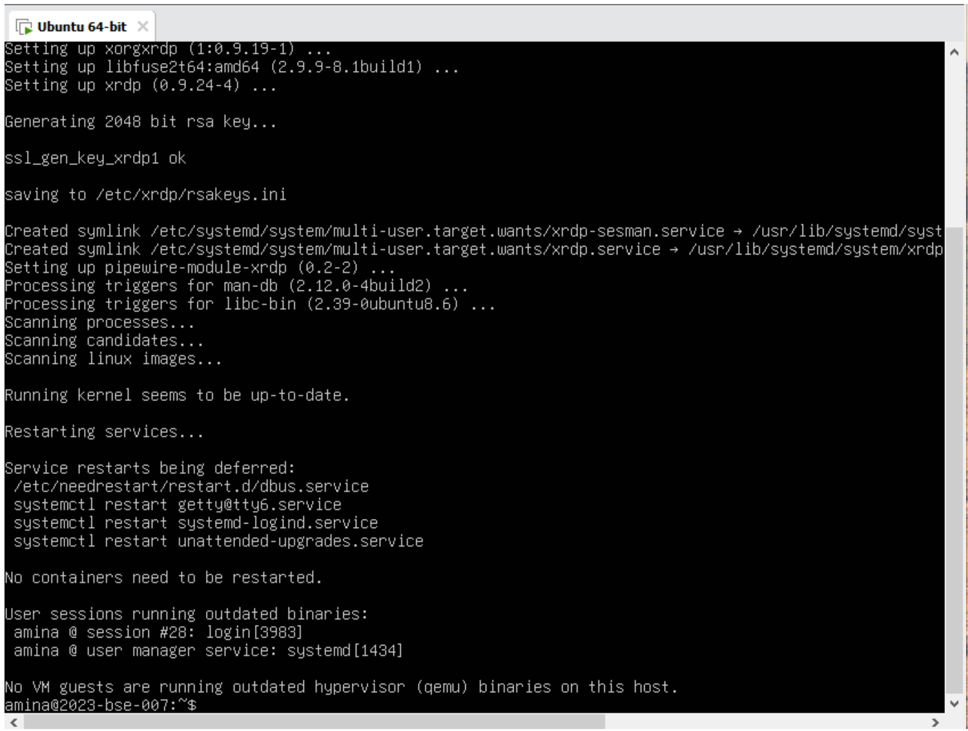
- This may take several minutes
- If asked about display manager, choose lightdm (if prompted)



Step 4: Install and enable XRDP

Install XRDP:

sudo apt install xrdp -y



Enable and start XRDP:

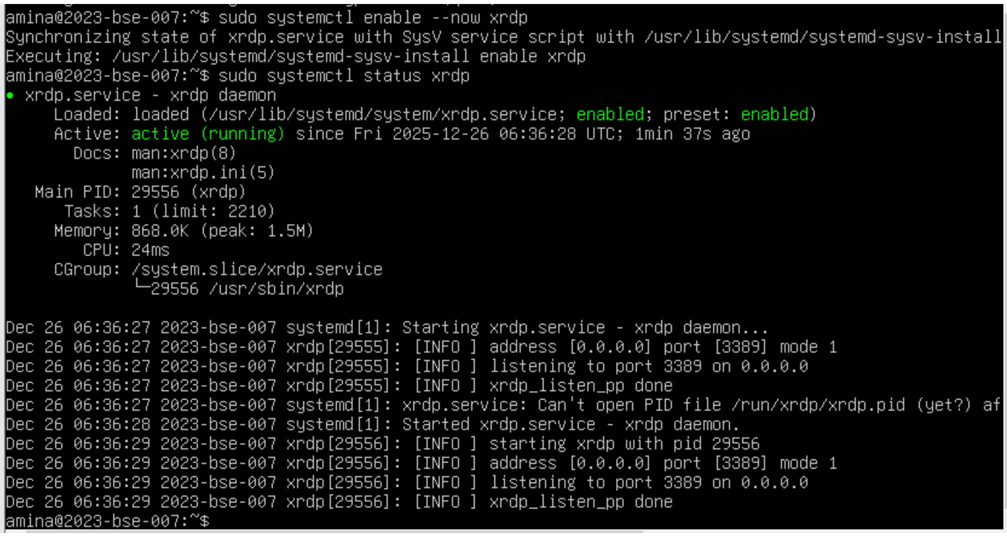
sudo systemctl enable --now xrdp

Step 5: Verify XRDP service status

Run:

sudo systemctl status xrdp

- Look for:
- Active: active (running)



Step 6: Configure XRDP to use XFCE

Run:

echo xfce4-session > ~/.xsession

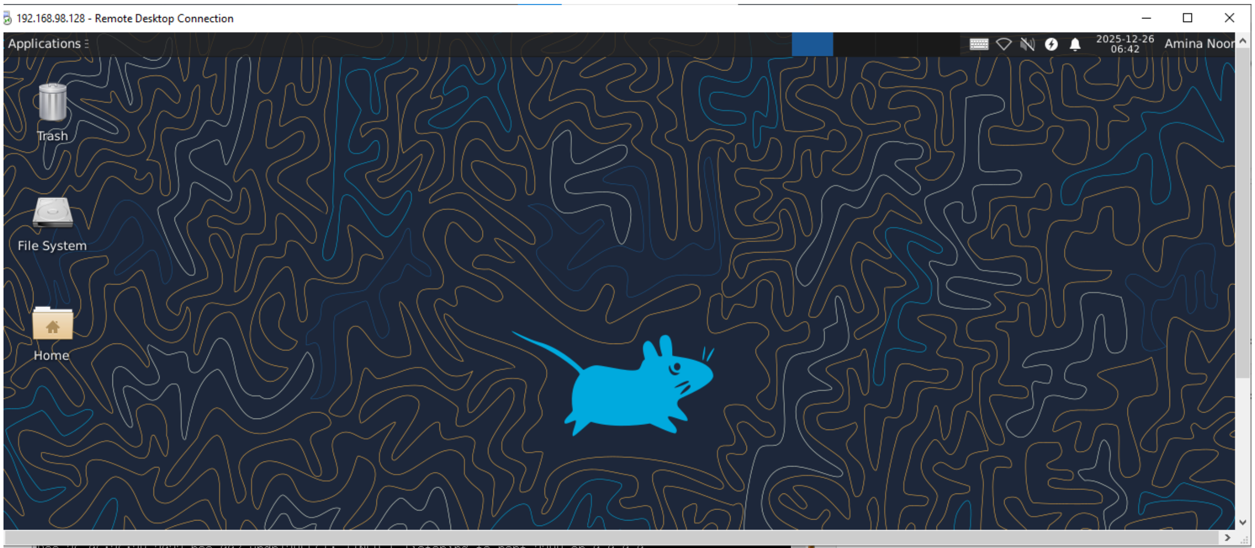
Verify:

cat ~/.xsession

Step 7: Connect using Remote Desktop (RDP)

From Windows

1. Press Win + R
2. Type:
3. mstsc
4. Enter your VM IP address
5. Login using:
    - Username: student
    - Password: your VM password

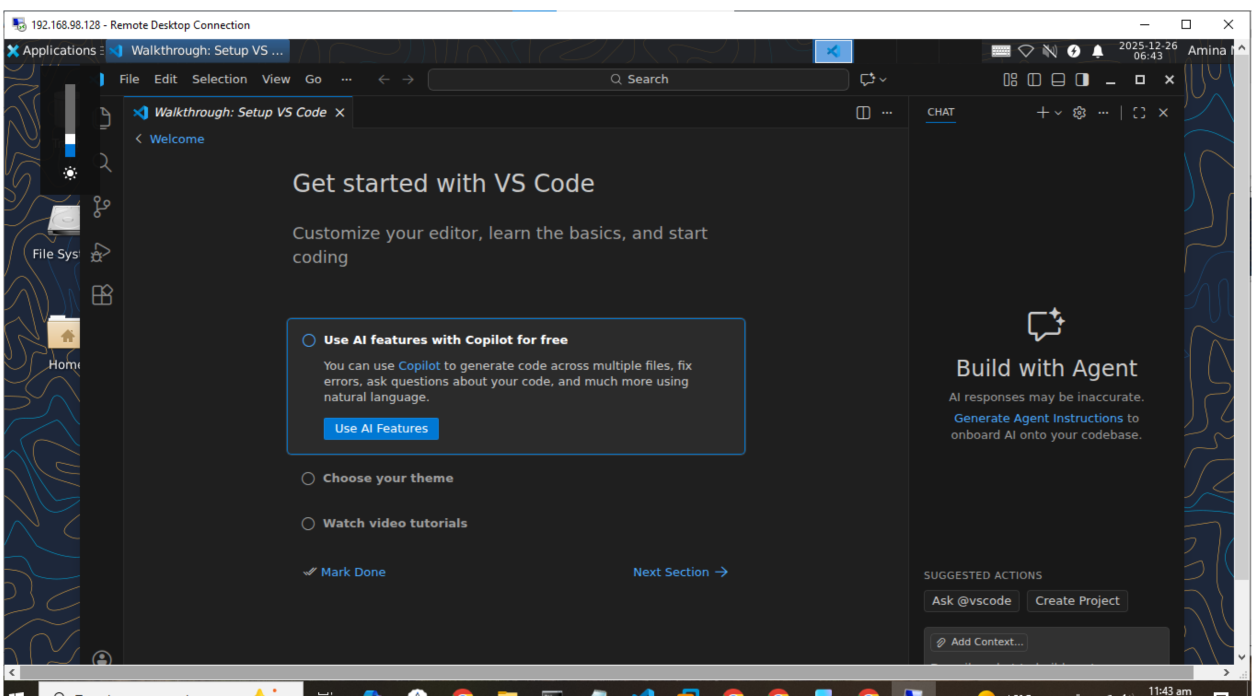


Step 8: Launch Visual Studio Code (GUI)

Inside the GUI desktop:

GUI menu

- Applications → Development → Visual Studio Code



Step 9: Logout & close session

- Log out from XFCE

- Close RDP
- Exit SSH

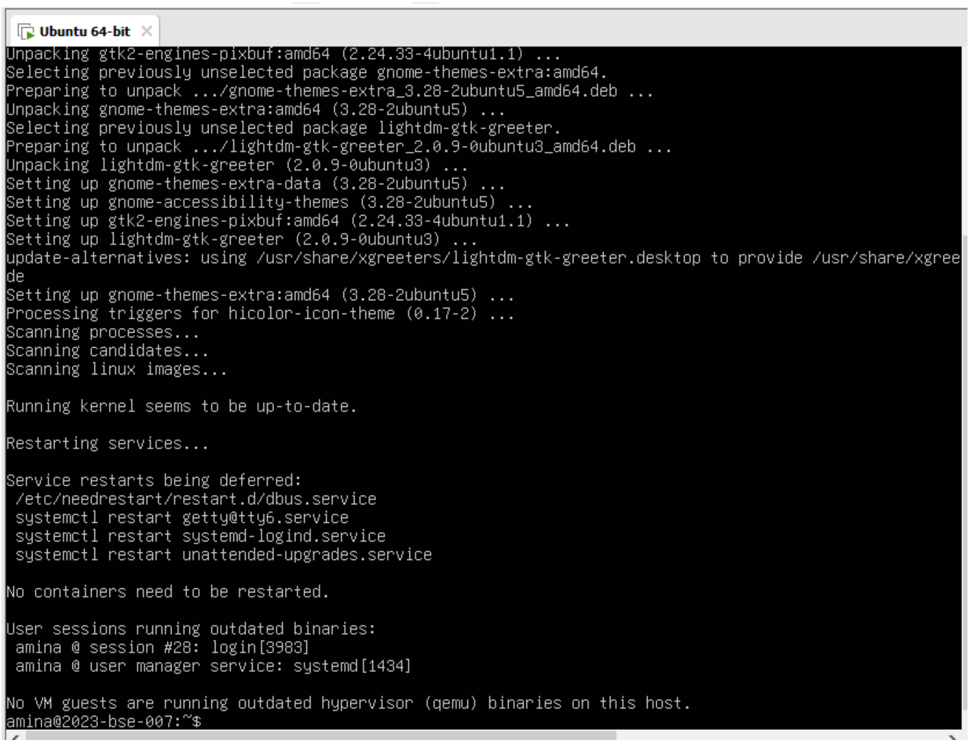
Task 6: LightDM + GUI Boot Control

PART 1: Install LightDM & GTK Greeter

Step 1: Install LightDM and greeter

Run **via SSH / terminal**:

sudo apt install lightdm lightdm-gtk-greeter -y




```
Unpacking gtk2-engines-pixbuf:amd64 (2.24.33-4ubuntu1.1) ...
Selecting previously unselected package gnome-themes-extra:amd64.
Preparing to unpack .../gnome-themes-extra_3.28-2ubuntu5_amd64.deb ...
Unpacking gnome-themes-extra:amd64 (3.28-2ubuntu5) ...
Selecting previously unselected package lightdm-gtk-greeter.
Preparing to unpack .../lightdm-gtk-greeter_2.0.9-0ubuntu3_amd64.deb ...
Unpacking lightdm-gtk-greeter (2.0.9-0ubuntu3) ...
Setting up gnome-themes-extra-data (3.28-2ubuntu5) ...
Setting up gnome-accessibility-themes (3.28-2ubuntu5) ...
Setting up gtk2-engines-pixbuf:amd64 (2.24.33-4ubuntu1.1) ...
Setting up lightdm-gtk-greeter (2.0.9-0ubuntu3) ...
update-alternatives: using /usr/share/xgreeters/lightdm-gtk-greeter.desktop to provide /usr/share/xgree
de
Setting up gnome-themes-extra:amd64 (3.28-2ubuntu5) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
 /etc/needrestart/restart.d/dbus.service
 systemctl restart getty@tty6.service
 systemctl restart systemd-logind.service
 systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
 amina @ session #28: login[3983]
 amina @ user manager service: systemd[1434]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
amina@2023-bse-007:~$
```

PART 2: Configure LightDM to use XFCE

Run these commands **exactly**:

sudo mkdir -p /etc/lightdm/lightdm.conf.d

echo -e "[Seat:*]\ngreeter-session=lightdm-gtk-greeter\nuser-session=xfce\nautologin-user-timeout=0" | sudo tee /etc/lightdm/lightdm.conf.d/99-xfce.conf

```
amina@2023-bse-007:~$ echo -e"[Seat:*]\ngreeter-session=lightdm-gtk-greeter\nuser-session=xfce\nautolog
e.conf
sudo: tree: command not found
amina@2023-bse-007:~$ _
```

PART 3: Clean session & permission problems

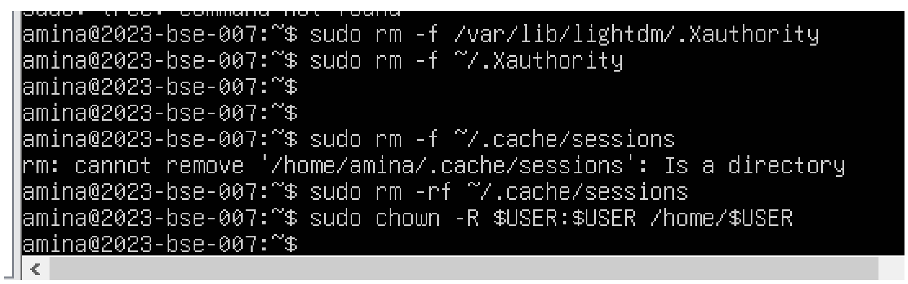
Run **each command**:

sudo rm -f /var/lib/lightdm/.Xauthority

sudo rm -f ~/.Xauthority

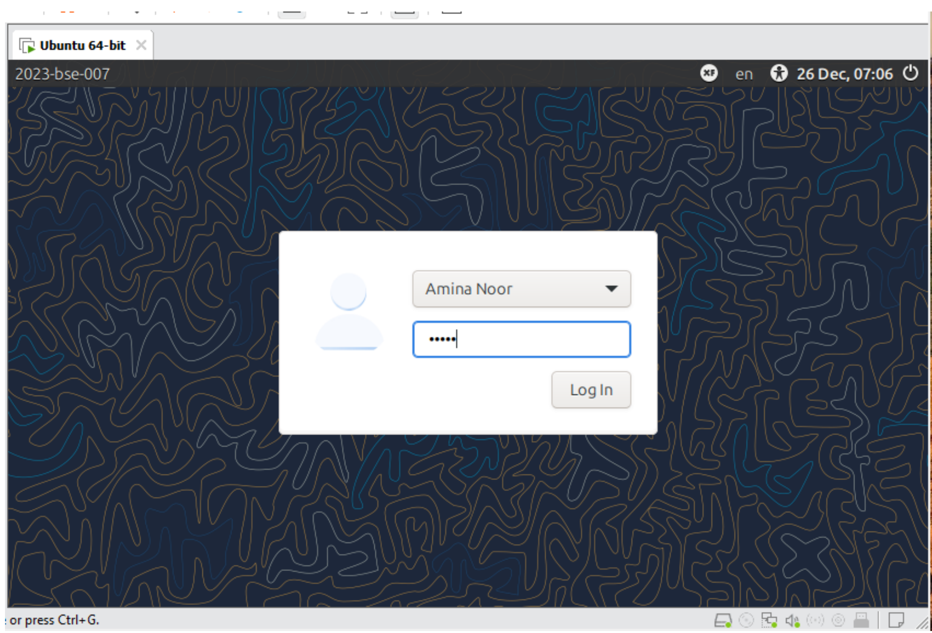
sudo rm -rf ~/.cache/sessions

sudo chown -R $USER:$USER /home/$USER



PART 4: Restart LightDM

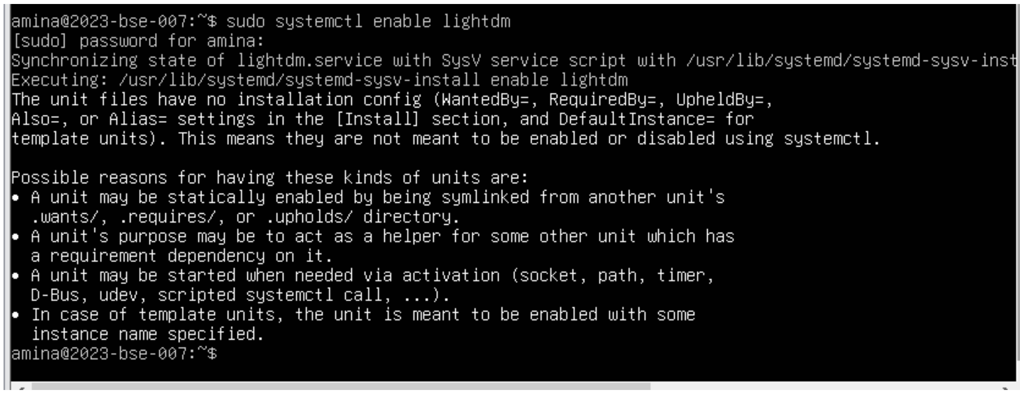
sudo systemctl restart lightdm



PART 5: ENABLE GUI at Boot (Graphical Target)

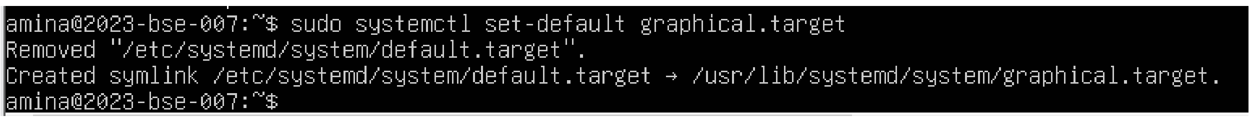
Step 5.1: Enable LightDM & graphical boot

sudo systemctl enable lightdm



```
amina@2023-bse-007:~$ sudo systemctl enable lightdm
[sudo] password for amina:
Synchronizing state of lightdm.service with SysV service script with /usr/lib/systemd/systemd-sysv-inst
Executing: /usr/lib/systemd/systemd-sysv-install enable lightdm
The unit files have no installation config (WantedBy=, RequiredBy=, UpheldBy=,
Also=, or Alias= settings in the [Install] section, and DefaultInstance= for
template units). This means they are not meant to be enabled or disabled using systemctl.

Possible reasons for having these kinds of units are:
• A unit may be statically enabled by being symlinked from another unit's
  .wants/, .requires/, or .upholds/ directory.
• A unit's purpose may be to act as a helper for some other unit which has
  a requirement dependency on it.
• A unit may be started when needed via activation (socket, path, timer,
  D-Bus, udev, scripted systemctl call, ...).
• In case of template units, the unit is meant to be enabled with some
  instance name specified.
amina@2023-bse-007:~$
```

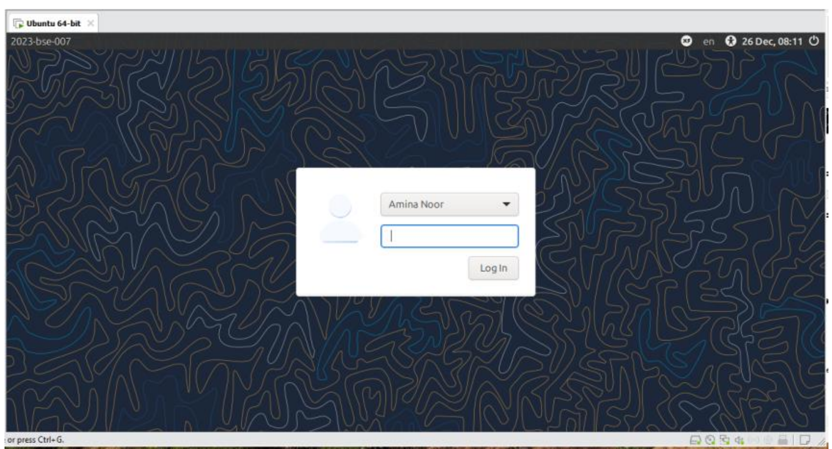
sudo systemctl set-default graphical.target



```
amina@2023-bse-007:~$ sudo systemctl set-default graphical.target
Removed "/etc/systemd/system/default.target".
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/graphical.target.
amina@2023-bse-007:~$
```

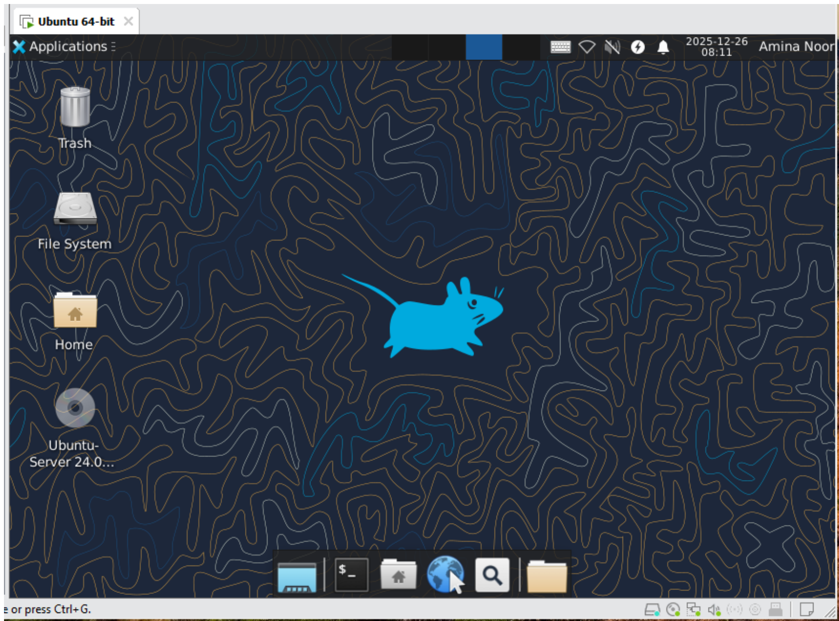
Step 5.2: Reboot to GUI

sudo reboot



After reboot:

- System should show **GUI login screen (LightDM GTK greeter)**
- Login with your **username + password**



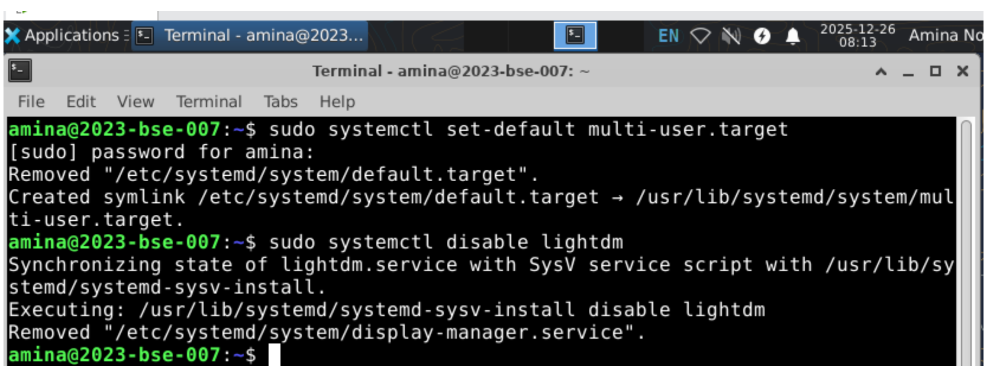
PART 6: DISABLE GUI at Boot

Step 6.1: Set CLI boot mode

After logging in (GUI or SSH):

sudo systemctl set-default multi-user.target

sudo systemctl disable lightdm

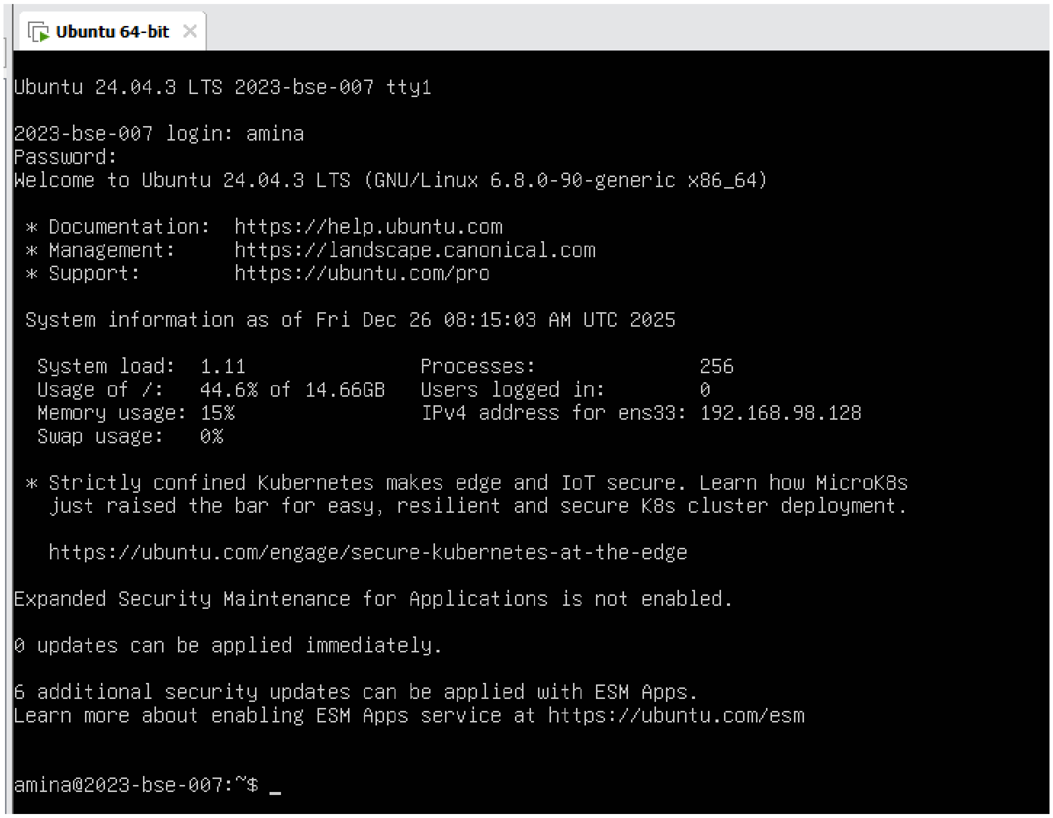


Step 6.2: Reboot to CLI

sudo reboot

After reboot:

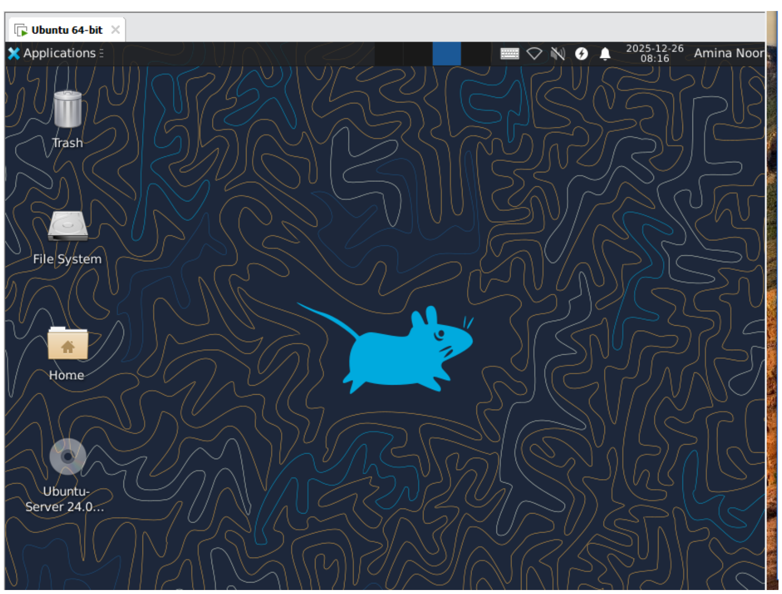
- You should see **text login prompt only**



PART 7: Start & Stop GUI Manually

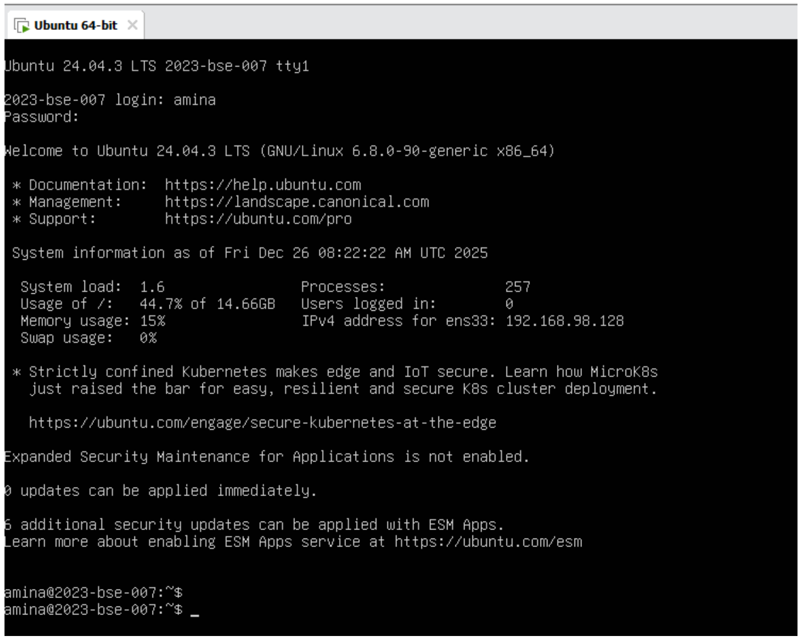
Step 7.1: Start GUI manually

sudo systemctl start lightdm



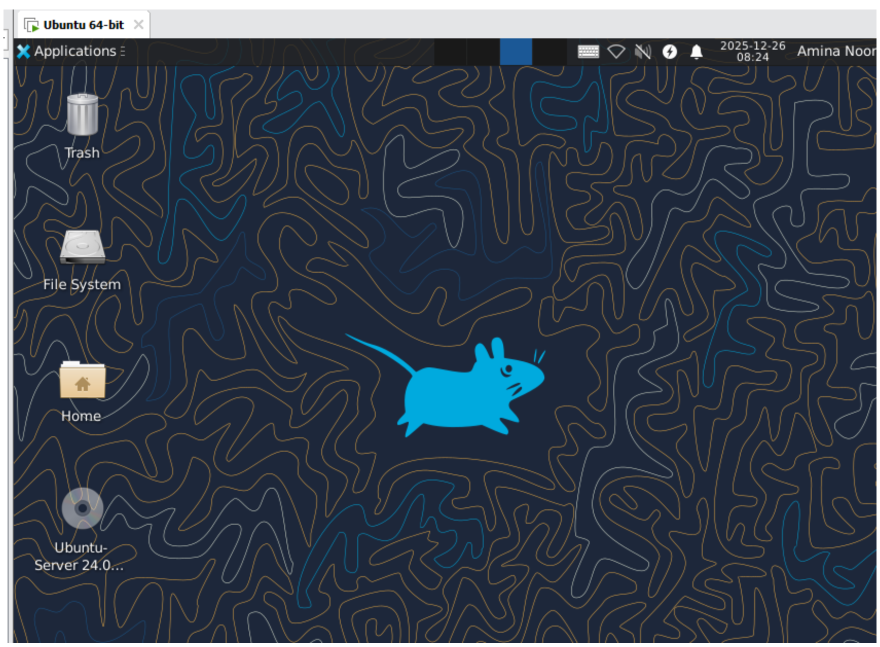
Step 7.2: Stop GUI manually

sudo systemctl stop lightdm



Step 7.3: Start GUI again (final start)

sudo systemctl start lightdm



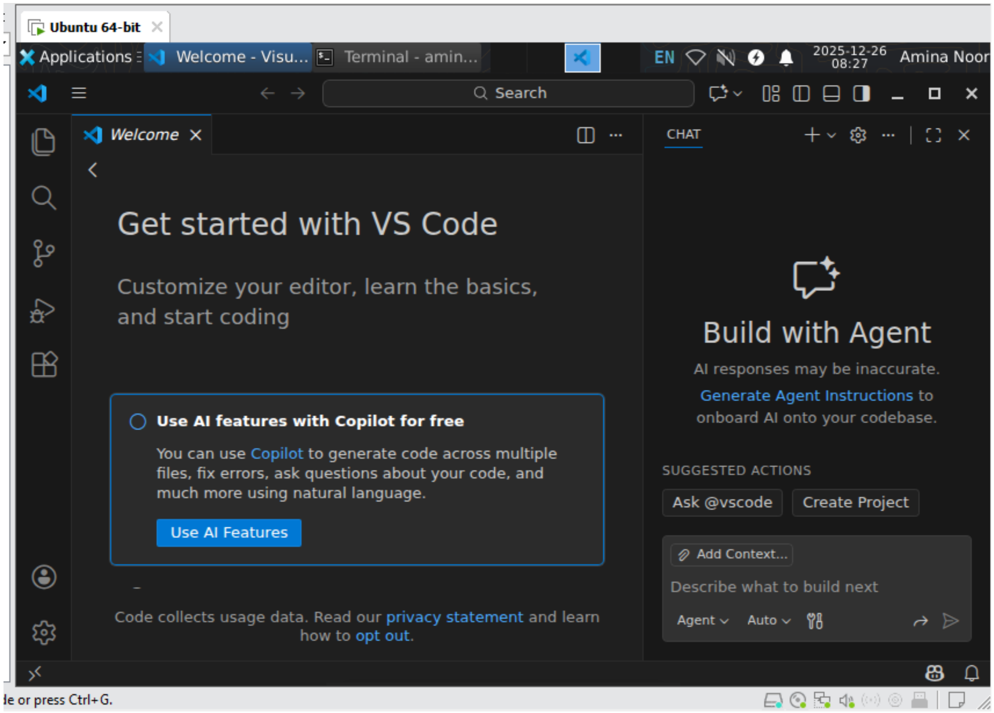
PART 8: Launch VS Code in GUI

Open **Terminal (inside GUI)**

Run:

code

VS Code window should open.



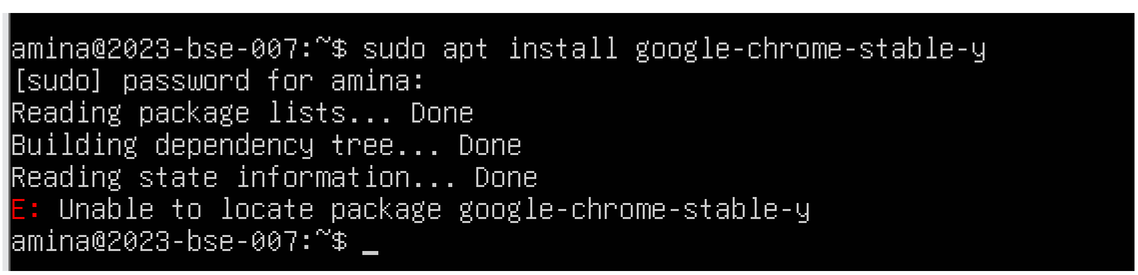
Close VS Code using ✕ button.

Task 7: Install Google Chrome by adding its apt source & key (Chrome)

Step 1: Attempt to install Chrome directly

We start by attempting installation **without the repository or key**. This is meant to fail.

sudo apt install google-chrome-stable -y



This failure happens because your system **doesn't know about Google's repo yet**.

Step 2: Inspect /etc/apt

Check the contents of the apt directory to see what's already there:

ls -la /etc/apt

```
amina@2023-bse-007:~$ ls -la /etc/apt
total 48
drwxr-xr-x   9 root root  4096 Dec 26 00:28 .
drwxr-xr-x 142 root root 12288 Dec 26 06:41 ..
drwxr-xr-x   2 root root  4096 Dec 26 05:27 apt.conf.d
drwxr-xr-x   2 root root  4096 Mar 31  2024 auth.conf.d
drwxr-xr-x   2 root root  4096 Mar 31  2024 keyrings
drwxr-xr-x   2 root root  4096 Dec 26 05:27 preferences.d
drwxr-xr-x   2 root root  4096 Aug  5 17:14 preferences.d.save
-rw-r--r--   1 root root    70 Dec 26 00:46 sources.list
drwxr-xr-x   2 root root  4096 Dec 26 00:46 sources.list.d
drwxr-xr-x   2 root root  4096 Aug  5 17:01 trusted.gpg.d
amina@2023-bse-007:~$
```

cat**Step 3: View main sources list**

The main sources list contains official repositories.

cat /etc/apt/sources.list

```
amina@2023-bse-007:~$ cat /etc/apt/sources.list
# Ubuntu sources have moved to /etc/apt/sources.list.d/ubuntu.sources
amina@2023-bse-007:~$
```

Step 4: List files under sources.list.d

This directory often has **additional third-party repositories**.

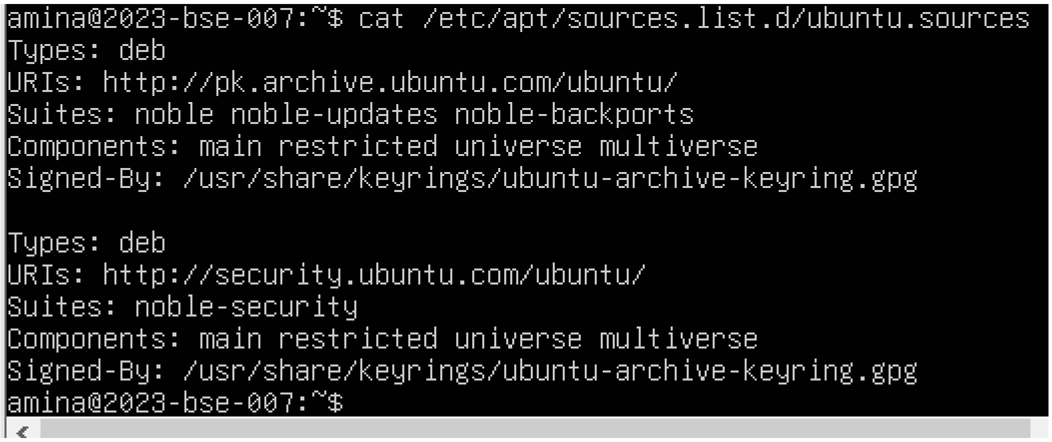
ls -la /etc/apt/sources.list.d/

```
amina@2023-bse-007:~$ ls -la /etc/apt/sources.list.d/
total 16
drwxr-xr-x 2 root root 4096 Dec 26 00:46 .
drwxr-xr-x 9 root root 4096 Dec 26 00:28 ..
-rw-r--r-- 1 root root  386 Dec 26 00:27 ubuntu.sources
-rw-r--r-- 1 root root 2552 Aug  5 17:02 ubuntu.sources.curtin.orig
amina@2023-bse-007:~$
```

Step 5: Check ubuntu.sources (if present)

Some systems have ubuntu.sources for third-party repos. If it exists:

cat /etc/apt/sources.list.d/ubuntu.sources



```
amina@2023-bse-007:~$ cat /etc/apt/sources.list.d/ubuntu.sources
Types: deb
URIs: http://pk.archive.ubuntu.com/ubuntu/
Suites: noble noble-updates noble-backports
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

Types: deb
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble-security
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
amina@2023-bse-007:~$
```

Step 6: Add Chrome repository

Open the file for editing:

sudo nano /etc/apt/sources.list.d/ubuntu.sources

Append these lines **at the end**:

Types: deb

URIs: http://dl.google.com/linux/chrome/deb/

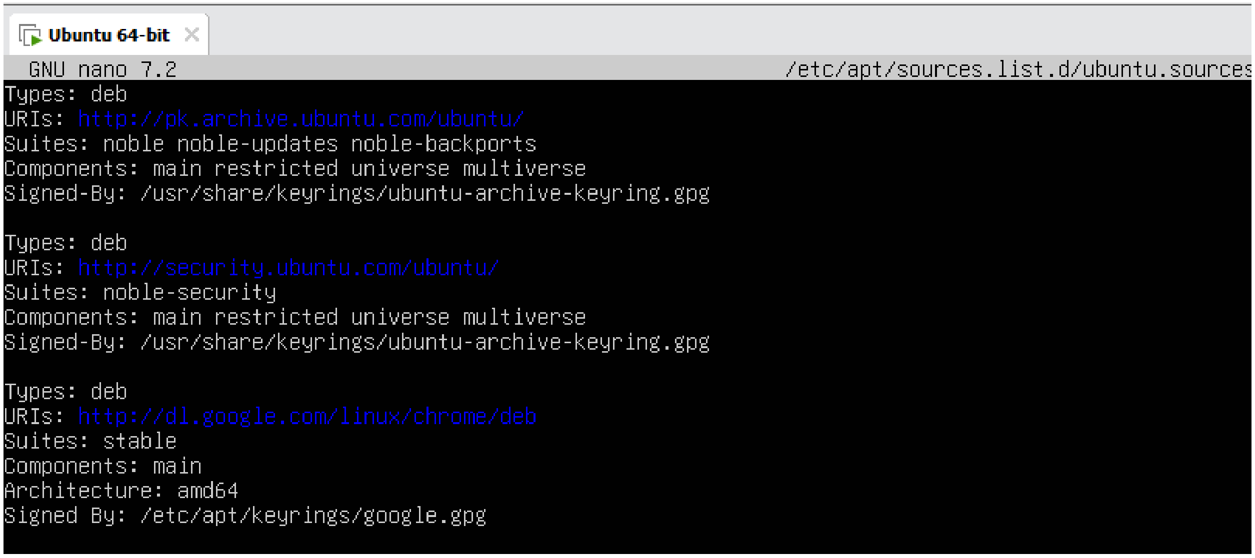
Suites: stable

Components: main

Architectures: amd64

Signed-By: /etc/apt/keyrings/google.gpg

Save and exit (Ctrl+O → Enter → Ctrl+X).



Step 7: Ensure keyrings folder exists and add Google's key

Create the keyrings folder:

sudo mkdir -p /etc/apt/keyrings

Download and add Google's signing key:

curl -fsSL https://dl.google.com/linux/linux_signing_key.pub | sudo gpg --dearmor -o /etc/apt/keyrings/google.gpg

Step 8: Update apt and install Chrome

Now that repo + key are added:

sudo apt update

```
amina@2023-bse-007:~$ sudo apt update
Get:1 http://dl.google.com/linux/chrome/deb stable InRelease [1,825 B]
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Err:1 http://dl.google.com/linux/chrome/deb stable InRelease
  The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 32E
Hit:3 http://pk.archive.ubuntu.com/ubuntu noble InRelease
Get:4 http://pk.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:5 http://pk.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
W: GPG error: http://dl.google.com/linux/chrome/deb stable InRelease: The following signatures couldn
PUBKEY 32EE5355A6BC6E42
E: The repository 'http://dl.google.com/linux/chrome/deb stable InRelease' is not signed.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
N: Missing Signed-By in the sources.list(5) entry for 'http://dl.google.com/linux/chrome/deb'
amina@2023-bse-007:~$ _
```

sudo apt install google-chrome-stable -y

```
amina@2023-bse-007:~$ sudo apt install google-chrome-stable -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package google-chrome-stable
```

Step 9: Add Chrome repository using one line

Create a dedicated list file for Chrome:

echo "deb [arch=amd64 signed-by=/etc/apt/keyrings/google.gpg] http://dl.google.com/linux/chrome/deb/ stable main" | sudo tee /etc/apt/sources.list.d/google-chrome.list > /dev/null

```
amina@2023-bse-007:~$ wget https://dl.google.com/linux/direct/google-chrome-stable_current_amd64.deb
--2026-01-10 05:40:26--  https://dl.google.com/linux/direct/google-chrome-stable_current_amd64.deb
Resolving dl.google.com (dl.google.com)... 142.250.186.238
Connecting to dl.google.com (dl.google.com)|142.250.186.238|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 117896052 (112M) [application/x-debian-package]
Saving to: 'google-chrome-stable_current_amd64.deb'

google-chrome-stable_current_amd64.deb  100%[==========================================================================>] 112.43M  1.16MB/s

2026-01-10 05:42:13 (1.06 MB/s) - 'google-chrome-stable_current_amd64.deb' saved [117896052/117896052]

amina@2023-bse-007:~$
```

Verify the file exists:

ls -la /etc/apt/sources.list.d/

```
amina@2023-bse-007:~$ ls -l /etc/apt/sources.list.d/
total 8
-rw-r--r-- 1 root root  388 Jan 10 05:28 ubuntu.sources
-rw-r--r-- 1 root root 2552 Aug  5 17:02 ubuntu.sources.curtin.orig
```

Step 10: Re-add the Google signing key (if removed)

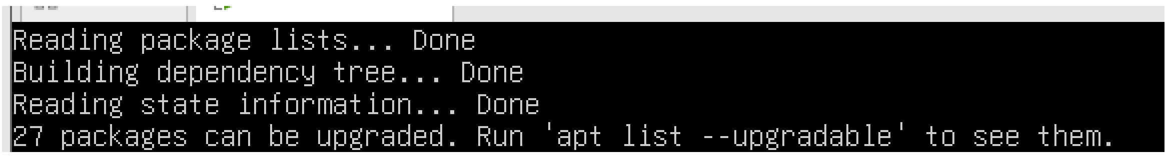
sudo mkdir -p /etc/apt/keyrings

curl -fsSL https://dl.google.com/linux/linux_signing_key.pub | sudo gpg --dearmor -o /etc/apt/keyrings/google.gpg

```
amina@2023-bse-007:~$ sudo mkdir -p /etc/apt/keyrings
amina@2023-bse-007:~$ curl -fsSL https://dl.google.com/linux/linux_signing_key.pub | sudo gpg --dearmor -o etc/apt/keyrings/google.gpg
gpg: can't create 'etc/apt/keyrings/google.gpg': No such file or directory
gpg: no valid OpenPGP data found.
gpg: dearmoring failed: No such file or directory
curl: (23) Failure writing output to destination
```

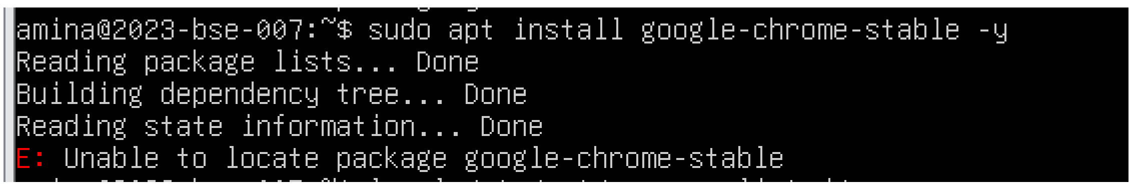
Step 12: Update apt and install Chrome (preferred flow)

sudo apt update



```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
27 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

sudo apt install google-chrome-stable -y



```
amina@2023-bse-007:~$ sudo apt install google-chrome-stable -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package google-chrome-stable
```

Task 8: Install Applications via PPA (Audacity & OBS)

Goal: Add PPAs → update apt → install → verify/launch → screenshots

PART A: Audacity via PPA

Step 1: Add Audacity PPA

Open terminal and run:

sudo add-apt-repository ppa:ubuntuhandbook1/audacity -y

```
amina@2023-bse-007:~$ sudo add-apt-repository ppa:ubuntuhandbook1/audacity -y
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu/
Suites: noble
Components: main
'
Description:
Unofficial build of Audacity audio editor

For help, please use Audacity forum: http://forum.audacityteam.org/

If the packages here are helpful, you may buy me a coffee:

        https://ko-fi.com/ubuntuhandbook1
More info: https://launchpad.net/~ubuntuhandbook1/+archive/ubuntu/audacity
Adding repository.
Hit:1 http://pk.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://pk.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://pk.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:5 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease [18.1 kB]
Get:6 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble/main amd64 Packages [1,0
Get:7 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble/main Translation-en [492
Fetched 19.6 kB in 3s (6,331 B/s)
Reading package lists... Done
amina@2023-bse-007:~$ _
```

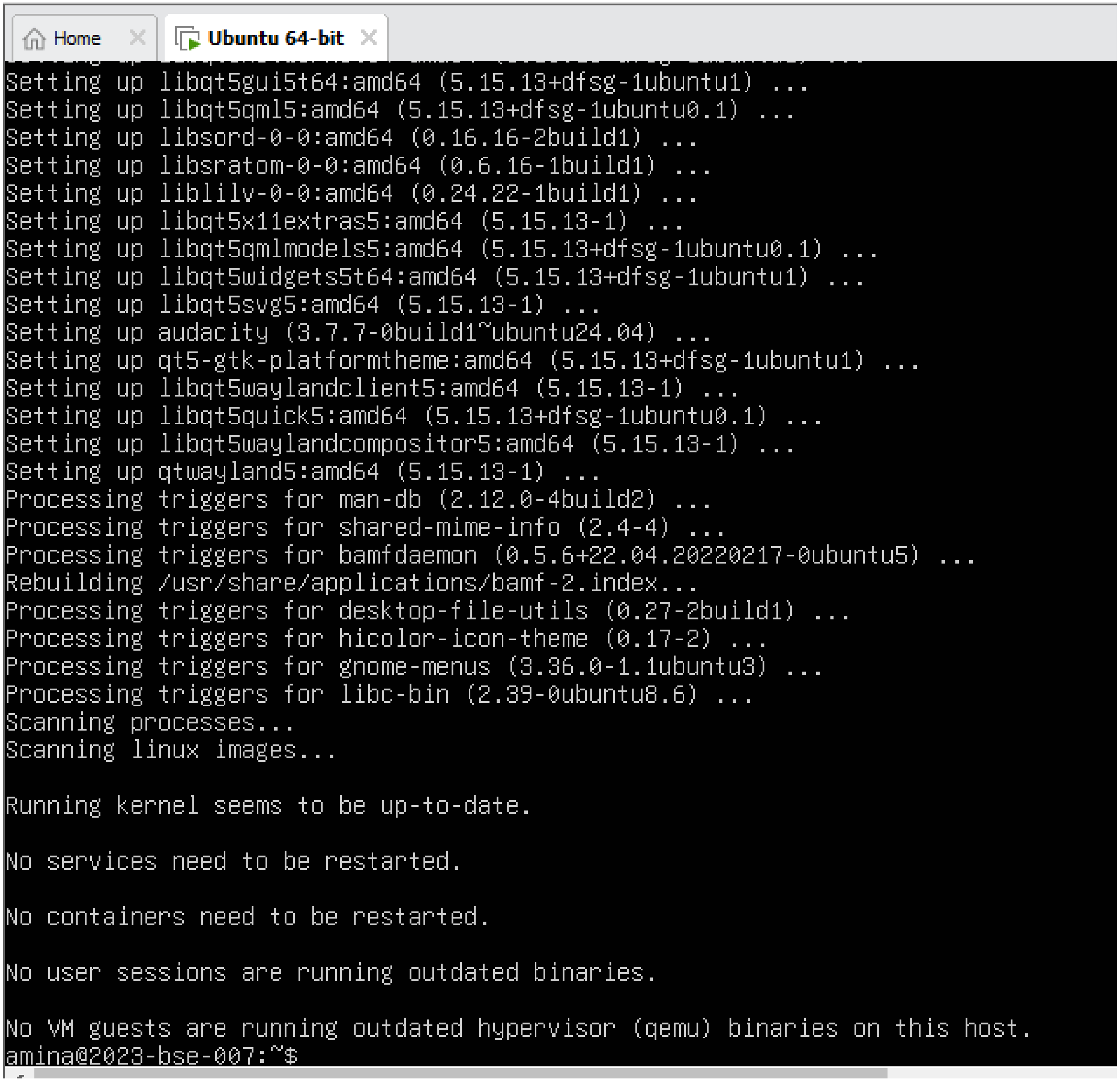
Step 2: Update apt

sudo apt update

```
amina@2023-bse-007:~$ sudo apt update
Hit:1 http://pk.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://pk.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 http://pk.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
27 packages can be upgraded. Run 'apt list --upgradable' to see them.
amina@2023-bse-007:~$
```

Step 3: Install Audacity

sudo apt install audacity -y



Step 4: Launch / Verify Audacity

audacity --version

```
amina@2023-bse-007:~$ apt list --installed | grep audacity

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

audacity-data/noble,now 3.7.7-0build1~ubuntu24.04 all [installed,automatic]
audacity/noble,now 3.7.7-0build1~ubuntu24.04 amd64 [installed]
amina@2023-bse-007:~$ sudo apt list --installed | grep audacity

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

audacity-data/noble,now 3.7.7-0build1~ubuntu24.04 all [installed,automatic]
audacity/noble,now 3.7.7-0build1~ubuntu24.04 amd64 [installed]
amina@2023-bse-007:~$ audacity
amina@2023-bse-007:~$ audacity_
```

PART B: OBS Studio via PPA

Step 5: Add OBS PPA

sudo add-apt-repository ppa:obsproject/obs-studio -y

```
amina@2023-bse-007:~$ sudo add-apt-repository ppa:obsproject/obs-studio -y
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu/
Suites: noble
Components: main
'
Description:
Latest stable release of OBS Studio
More info: https://launchpad.net/~obsproject/+archive/ubuntu/obs-studio
Adding repository.
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://pk.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://pk.archive.ubuntu.com/ubuntu noble-updates InRelease
Get:4 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble InRelease [17.8 kB]
Hit:5 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Get:6 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble/main amd64 Packages [1,172
Get:7 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble/main Translation-en [160 B]
Hit:8 http://pk.archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 19.1 kB in 3s (6,579 B/s)
Reading package lists... Done
amina@2023-bse-007:~$
```

Step 6: Update apt

sudo apt update

```
amina@2023-bse-007:~$ sudo apt update
Hit:1 http://pk.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://pk.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble InRelease
Hit:5 http://pk.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:6 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
27 packages can be upgraded. Run 'apt list --upgradable' to see them.
amina@2023-bse-007:~$ _
```

Step 7: Install OBS Studio

sudo apt install obs-studio -y

```
Setting up libsphinxbase3t64:amd64 (0.8+5prealpha+1-17build2) ...
Setting up libqt6svg6:amd64 (6.4.2-4ubuntu3) ...
Setting up libqt6waylandclient6:amd64 (6.4.2-5build3) ...
Setting up libass9:amd64 (1:0.17.1-2build1) ...
Setting up qt6-gtk-platformtheme:amd64 (6.4.2+dfsg-21.1build5) ...
Setting up libqt6quick6:amd64 (6.4.2+dfsg-4build3) ...
Setting up libmbedtls14t64:amd64 (2.28.8-1) ...
Setting up libqt6wlshellintegration6:amd64 (6.4.2-5build3) ...
Setting up libqt6waylandcompositor6:amd64 (6.4.2-5build3) ...
Setting up libqt6waylandeglcompositorhwintegration6:amd64 (6.4.2-5build3) ...
Setting up libqt6waylandeglclienthwintegration6:amd64 (6.4.2-5build3) ...
Setting up libpocketsphinx3:amd64 (0.8.0+real5prealpha+1-15ubuntu5) ...
Setting up qt6-wayland:amd64 (6.4.2-5build3) ...
Setting up libavfilter9:amd64 (7:6.1.1-3ubuntu5) ...
Setting up libavdevice60:amd64 (7:6.1.1-3ubuntu5) ...
Setting up obs-studio (32.0.2-0obsproject1~noble) ...
Processing triggers for bamfdaemon (0.5.6+22.04.20220217-0ubuntu5) ...
Rebuilding /usr/share/applications/bamf-2.index...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for gnome-menus (3.36.0-1.1ubuntu3) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
amina@2023-bse-007:~$
```

Step 8: Launch / Verify OBS

obs

```
amina@2023-bse-007:~$ obs
qt.qpa.xcb: could not connect to display
qt.qpa.plugin: Could not load the Qt platform plugin "xcb" in "" even though it was found.
This application failed to start because no Qt platform plugin could be initialized. Reinstalling the applicatio

Available platform plugins are: eglfs, wayland, xcb, wayland-egl, minimalegl, minimal, linuxfb, vkkhrdisplay, vn

Aborted (core dumped)
amina@2023-bse-007:~$
```

Option B (CLI verification)

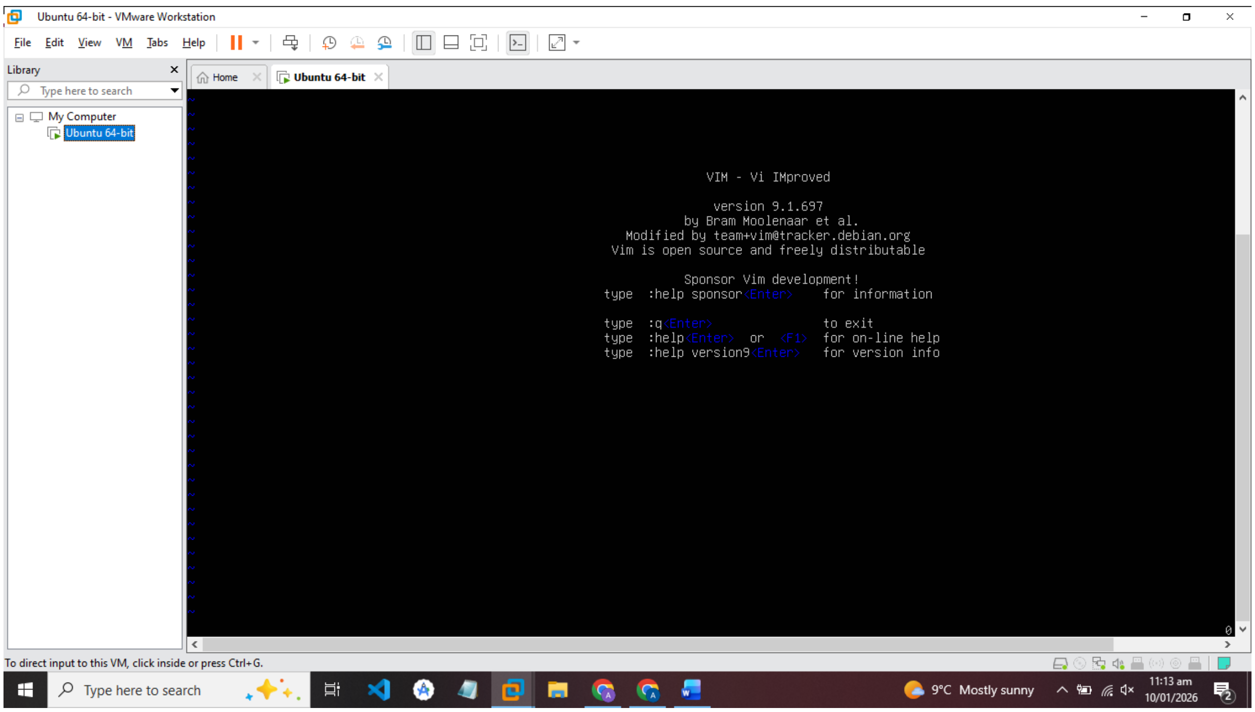
obs --version

Task 9: Create a Kubernetes sample YAML using vim

Step 1: Check if vim is installed

Run this command in your terminal:

Vim



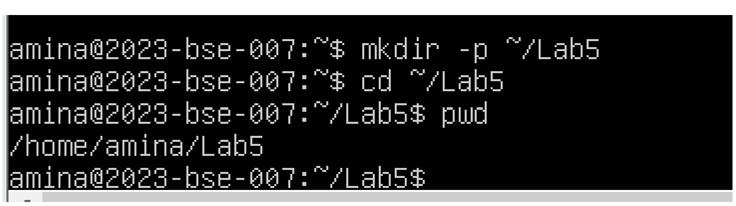
Step 2: Create Lab5 directory

Run:

mkdir -p ~/Lab5

cd ~/Lab5

pwd



Step 3: Create the Kubernetes YAML file

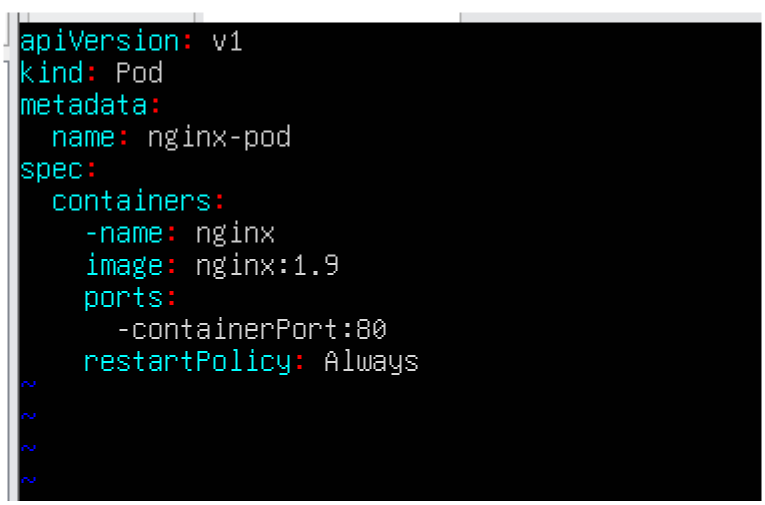
Run:

vim k8s-sample.yaml

- Vim opens a blank file.
- Press i to enter **insert mode**.

Paste the following YAML exactly (mind spaces and indentation—they are important!):

```
apiVersion: v1

kind: Pod

metadata:

  name: nginx-pod

spec:

  containers:

  - name: nginx

    image: nginx:1.19

    ports:

    - containerPort: 80

  restartPolicy: Always
```



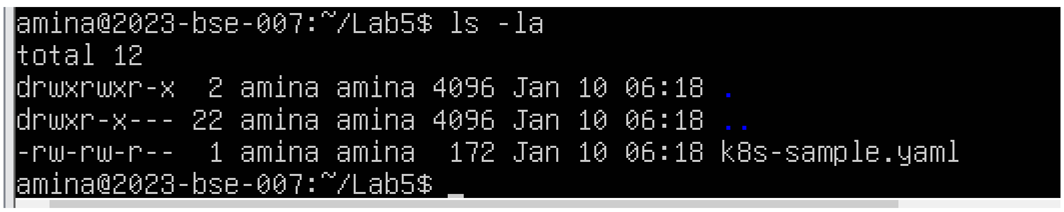
Step 4: Save and exit Vim

1. Press Esc to leave insert mode.
2. Type :wq and press Enter → this saves the file and exits vim.

Step 5: Verify the file was created

Run:

ls -la



```
amina@2023-bse-007:~/Lab5$ ls -la
total 12
drwxrwxr-x  2 amina amina 4096 Jan 10 06:18 .
drwxr-x--- 22 amina amina 4096 Jan 10 06:18 ..
-rw-rw-r--  1 amina amina  172 Jan 10 06:18 k8s-sample.yaml
amina@2023-bse-007:~/Lab5$ _
```

Task 10: Edit the Kubernetes YAML - add annotation, verify, then discard temporary change

Step 1: Open the YAML file in vim

cd ~/Lab5

vim k8s-sample.yaml

- Make sure you are in **command mode** by pressing Esc.
- Then press i to enter **insert mode**.

Step 2: Add an annotation

Add the following lines **under metadata:**, keeping proper indentation (2 spaces):

annotations:

  lab: lesson11

- After adding, press Esc to go back to **command mode**.
- Save and quit with:

:wq

- Verify the annotation by running:

cat k8s-sample.yaml

```
amina@2023-bse-007:~/Lab5$ cat k8s-sample.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    -name: nginx
    image: nginx:1.9
    ports:
      -containerPort:80
    restartPolicy: Always
amina@2023-bse-007:~/Lab5$ _
```

Step 3: Practice temporary edit and discard

1. Open the file again:

vim k8s-sample.yaml

2. Enter insert mode (i) and add a temporary comment **anywhere**:

temp: do-not-keep

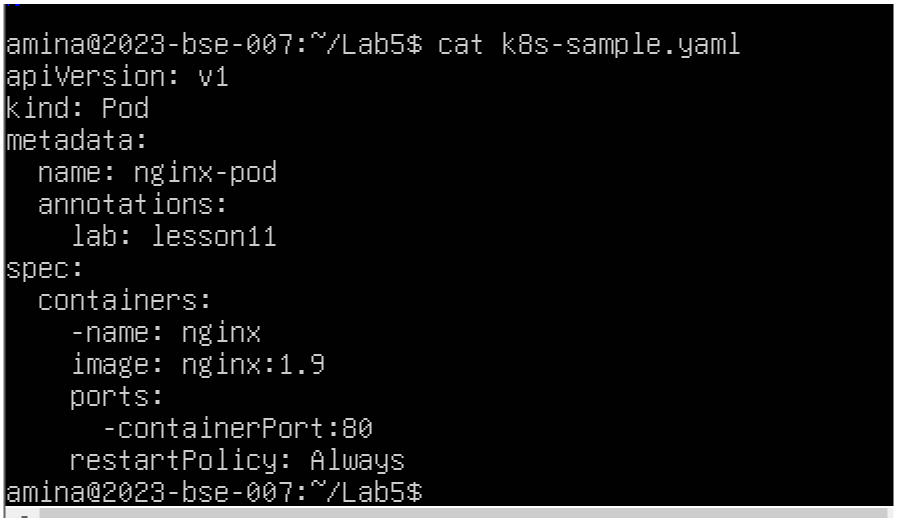
```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    -name: nginx
    image: nginx:1.9
    ports:
      -containerPort:80
    restartPolicy: Always
  #temp: do-not-keep
~
~
~
```

3. **Do NOT save**. Press Esc to go back to command mode, then **force quit without saving**:

:q!

Step 4: Verify the temporary change is gone

cat k8s-sample.yaml



Task 11: Vim editing practice - delete, undo, numeric deletes, navigation

Step 1: Open the file

cd ~/Lab5

vim k8s-sample.yaml

- Make sure you are in **command mode** (press Esc).

Step 2: Delete a single line (dd)

1. Move the cursor to the line containing:

image: nginx:1.19

2. Delete it with:

dd

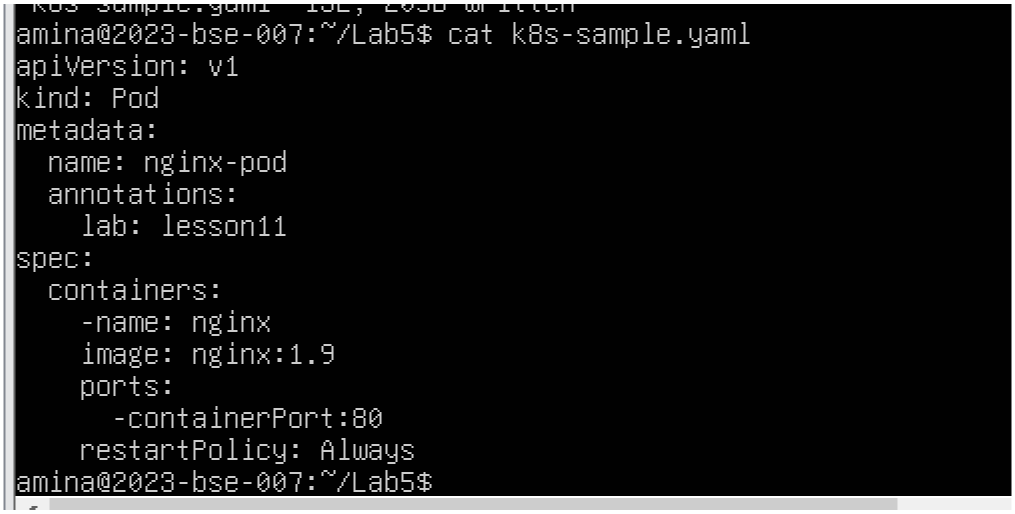
3. Immediately undo the deletion with:

u

4. Save and quit to capture file content for screenshot:

:wq

cat k8s-sample.yaml



```
amina@2023-bse-007:~/Lab5$ cat k8s-sample.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    -name: nginx
    image: nginx:1.9
    ports:
      -containerPort:80
    restartPolicy: Always
amina@2023-bse-007:~/Lab5$
```

Step 3: Delete multiple lines (numeric delete)

1. Open the file again:

vim k8s-sample.yaml

2. Move cursor to the **first line of the three you want to delete** (e.g., image: line).

3. Delete **3 lines at once** using numeric delete:

3dd

- Alternative: d3d also works. Document which one you used.

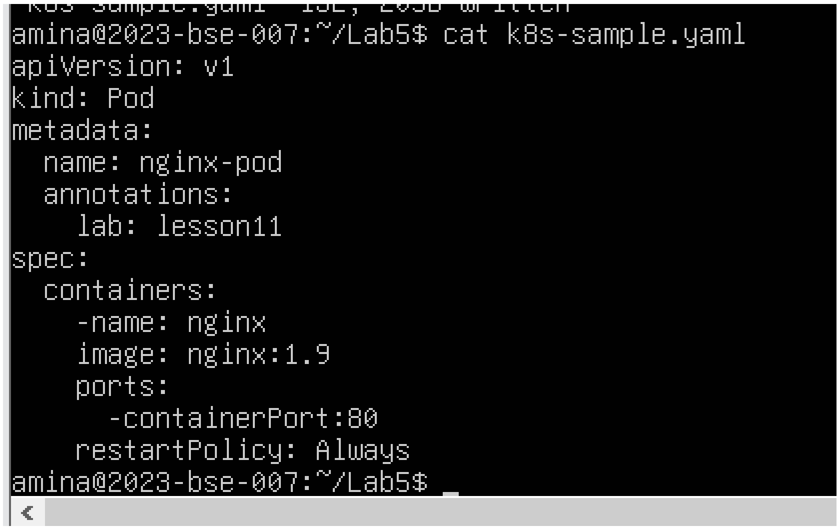
4. Immediately undo the deletion:

u

5. Save and quit, then verify with cat:

:wq

cat k8s-sample.yaml



Step 4: Navigation practice

Open the file again:

vim k8s-sample.yaml

- **Jump to the first line**:

1G

- Take a screenshot of the first line → save as: task11_line1.png

- **Jump to the last line**:

G

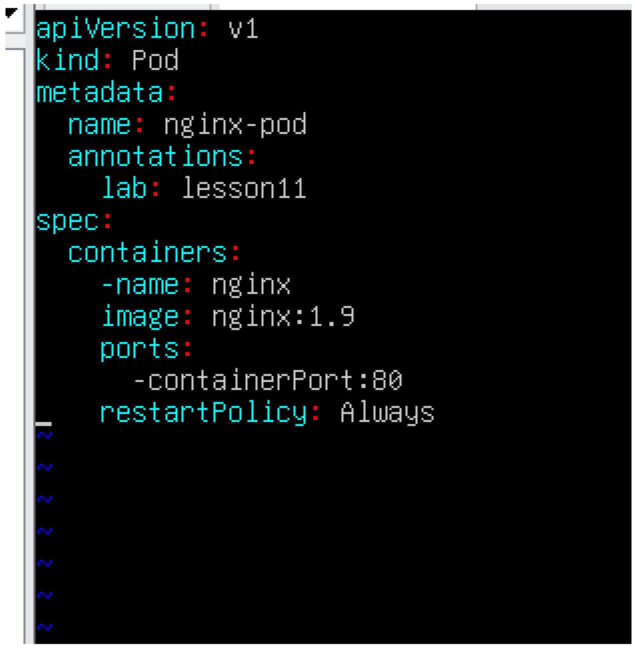
- Move cursor to the end of line containing containerPort: 80:

$

- Move back to start of line:

0

- Capture a screenshot showing cursor at the line and the commands



```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    -name: nginx
    image: nginx:1.9
    ports:
      -containerPort:80
    restartPolicy: Always
~
~
~
~
~
~
~
```

Step 5: Exit vim

:q

Task 12: Vim search, add matches, substitute, undo

Step 1: Open the file

cd ~/Lab5

vim k8s-sample.yaml

- Make sure you are in **command mode** (Esc).

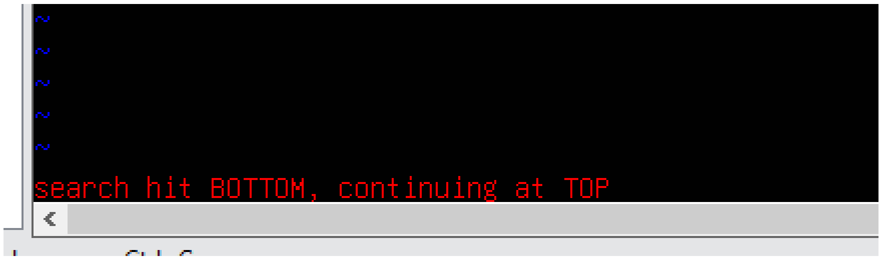
Step 2: Search for nginx

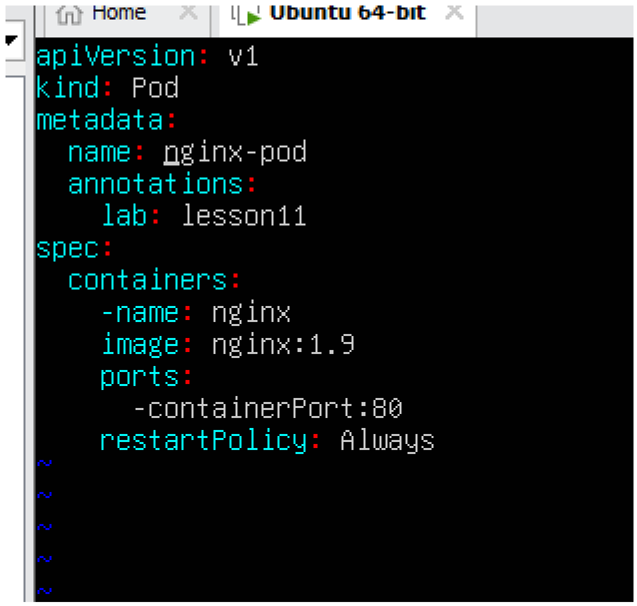
1. In command mode, type:

/nginx

2. Press Enter.

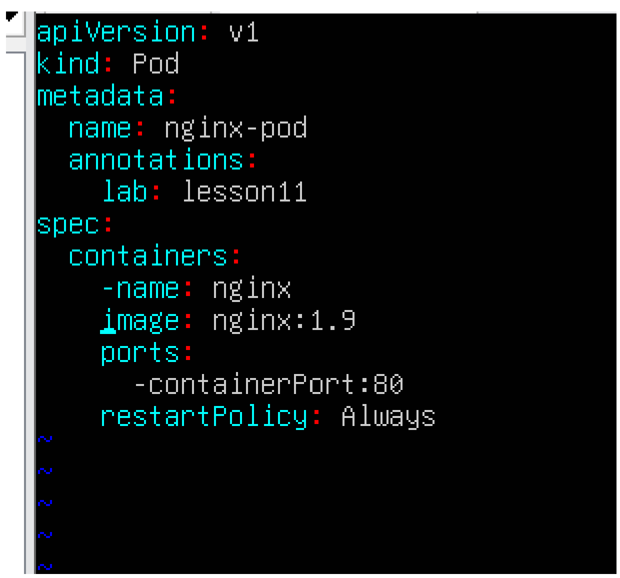
- The cursor will jump to the first occurrence of nginx and highlight it.





Step 3: Navigate matches with n and N

- Press n → goes to the **next match**
- Press N → goes **back to previous match**



Step 4: Add two more occurrences of nginx

1. Enter insert mode:

i

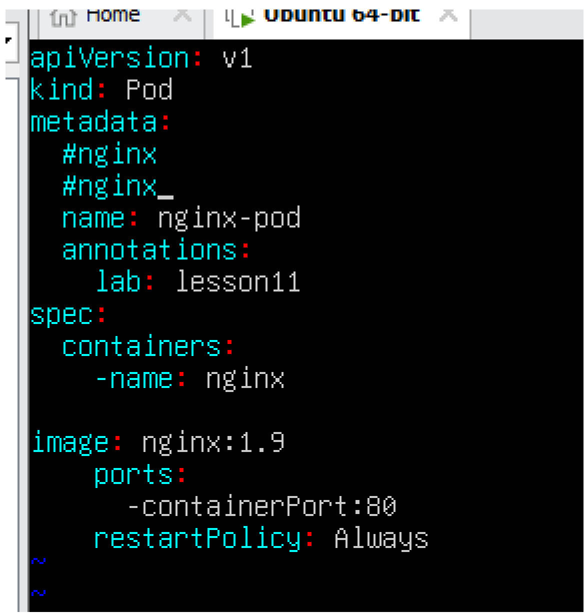
2. Add two occurrences of nginx (for example, as comments or under metadata):

nginx

nginx

3. Press Esc to return to command mode.
4. Save the file:

:w

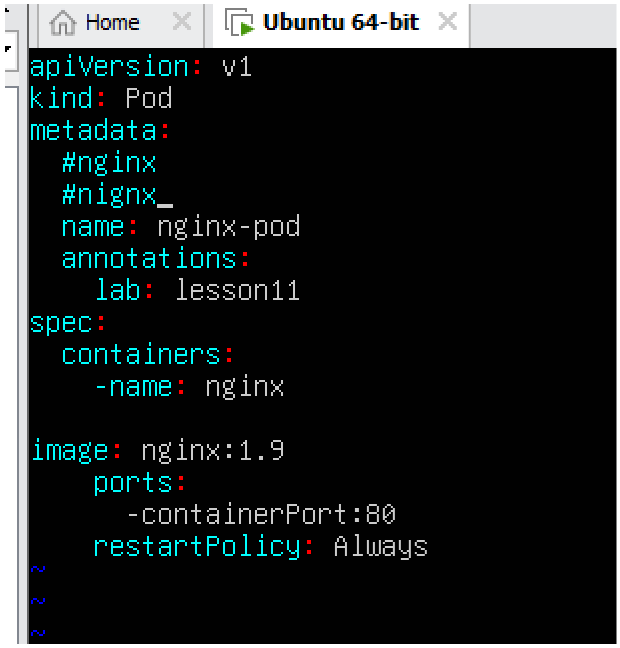


Step 5: Cycle through matches

1. From command mode, search again:

/nginx

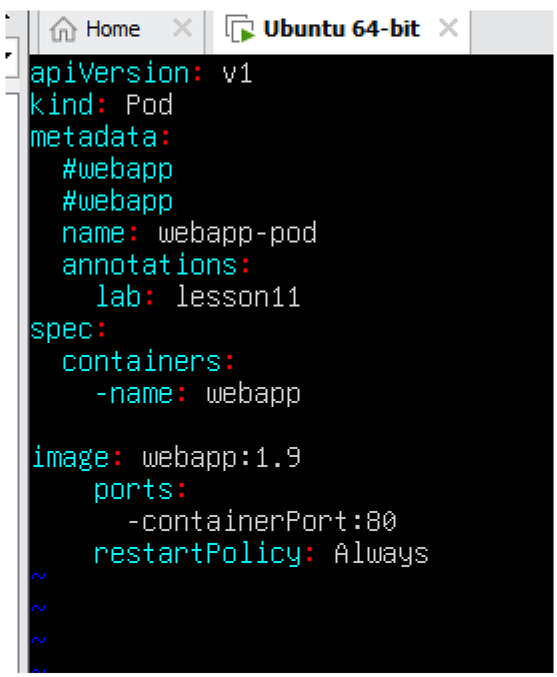
2. Press n repeatedly to cycle forward through **all occurrences**, including the ones you just added.



Step 6: Substitute all occurrences

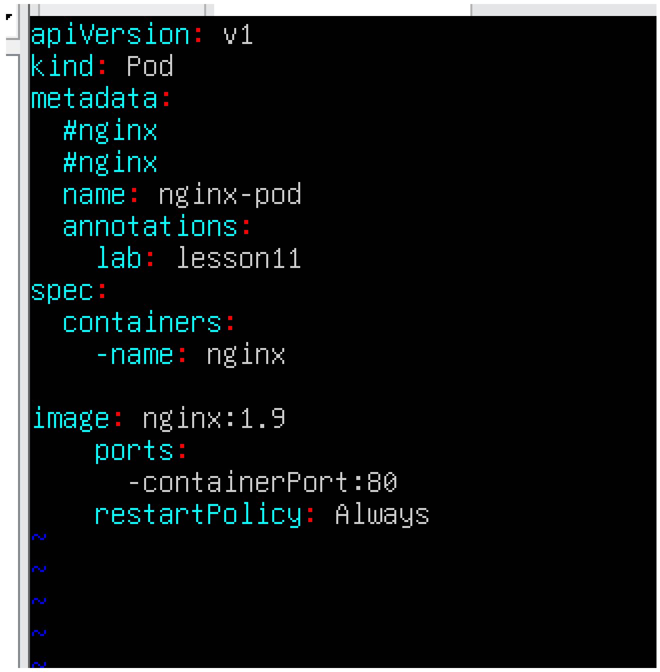
- From command mode, replace all nginx with webapp:

:%s/nginx/webapp



Step 7: Undo the substitution

- Press u in command mode → restores the file to previous state.



Step 8: Exit vim

- If you want to **discard any accidental changes**, exit without saving:

:q!

- If all changes are saved and undone correctly, exit normally:

:q

Exam Evaluation Question

Step 1: Understand the requirements

Docker Desktop is usually designed for **Ubuntu Desktop**, not server-only environments, because it needs a GUI to launch the Docker Desktop application.

Options on Ubuntu Server:

1. Install a **GUI desktop environment** (like XFCE, GNOME, or KDE) on the server so Docker Desktop can run.
2. Alternatively, you could install **Docker Engine CLI only**, but that's not Docker Desktop. For this task, you need the **GUI running Docker Desktop**.

Step 2: Prepare your Ubuntu VM

- Make sure your Ubuntu VM has:
    - Enough **RAM (≥ 4GB recommended)**
    - Enough **disk space (≥ 20GB)**
    - A **GUI installed** (XFCE or GNOME).
- Verify you can open GUI apps (like gedit or firefox) to ensure Docker Desktop will work.

Step 3: Research Docker Desktop for Linux

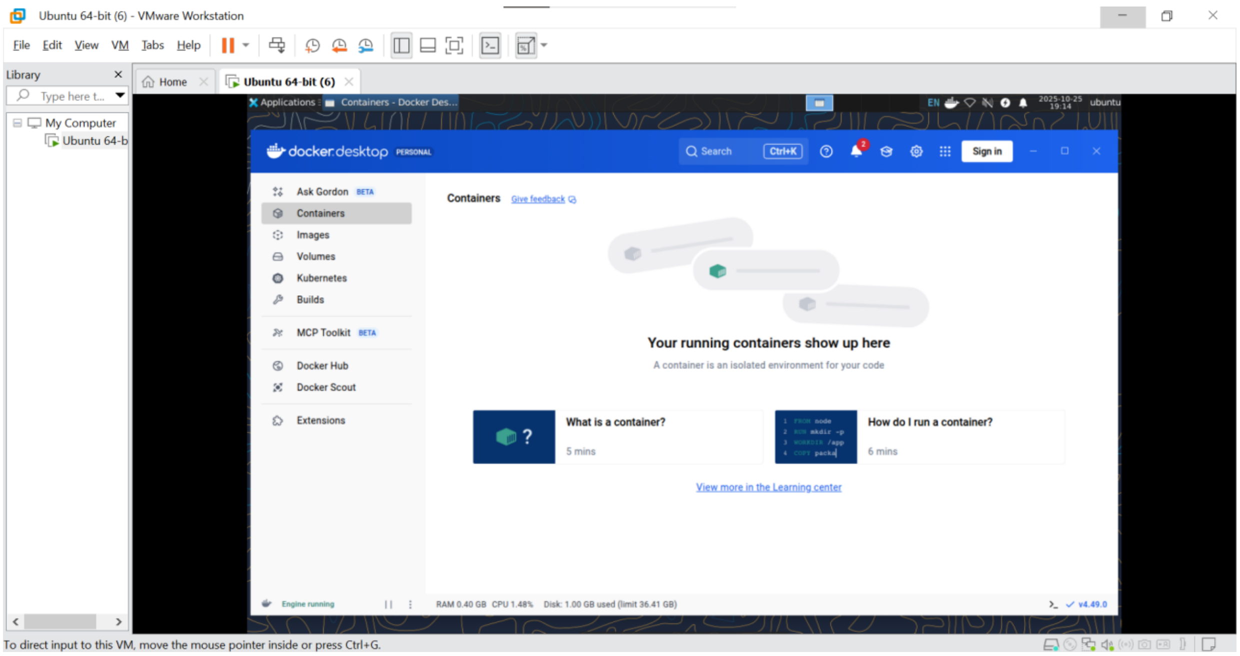
- Go to the official Docker Desktop Linux page: https://www.docker.com/products/docker-desktop/
- Check the **Ubuntu/Linux installation instructions**.
- You will find the .deb package or instructions to add the Docker repository, install Docker Desktop, and register your user to the Docker group.

Step 4: Install Docker Desktop

- Download the .deb package for Docker Desktop on Linux.
- Install it using the **package manager** (dpkg or GUI software installer).
- Add your user to the **docker group** so you can run Docker commands without sudo.
- Start Docker Desktop using the **GUI menu** or command line (docker-desktop), depending on your environment.

Step 5: Verify installation

- Launch Docker Desktop from the GUI (menu or application launcher).
- Check that it **starts correctly** (shows the dashboard).



Ubuntu 64-bit (6) - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Home
Ubuntu 64-bit (6)
Applications
Containers - Docker Des...
docker desktop PERSONAL
Search
Ctrl+K
Sign in
Ask Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Containers
Give feedback
Your running containers show up here
A container is an isolated environment for your code
What is a container?
5 mins
How do I run a container?
6 mins
View more in the Learning center
Engine running
RAM 0.40 GB CPU 1.48% Disk: 1.00 GB used (limit 36.41 GB)
v4.49.0
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.